DO MEMORY CONSTRAINTS AND DEVELOPMENTAL DATA MAKE LANGUAGE MODELS BETTER PREDICTORS OF HUMAN BEHAVIORAL AND NEURAL RESPONSES?

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ABSTRACT

Do language models and human brains arrive at similar solutions to language processing? Two lines of work aim to make models more human-like: architectural inductive biases mirroring human cognitive limits, such as working-memory bottlenecks on attention, and training restricted to the linguistic input a child receives, as in the BabyLM Challenge. We evaluate whether models constrained by memory, data, or both predict human data better than unconstrained ones. We introduce ***HumanScore-Language***, a benchmark measuring how accurately model-derived surprisal and contextual embeddings predict reading times and eye movements alongside brain activity (fMRI, MEG, EEG, N400 event-related potentials, intracranial recordings) across eighteen datasets, each normalized by its own noise ceiling. Evaluating 118 models in a 2×2 factorial design, with mixed models that account for the dependence between related models and between datasets, we find that the two constraints are associated with behavioral and neural prediction in opposite directions for contextual embeddings. At matched size and perplexity, a developmental data budget is associated with better prediction of reading behavior, by about one between-model standard deviation, but not of neural responses, whereas memory constraints are associated with worse neural prediction and no reliable behavioral gain. Combining both constraints cancels the behavioral advantage, and within memory-limited architectures less training data does not improve behavioral prediction. Matched comparisons locate the neural penalty in forget-gate and distance-biased attention. These domain differences are tested directly and are robust to resampling model families, excluding datasets, and outlier treatment. Surprisal-based results are weaker and depend on datasets with unstable noise ceilings. Larger models predict neural responses better from embeddings, whereas at matched perplexity they predict reading behavior worse from surprisal. Complementary representational analyses show that small and constrained models remain competitive on behavioral measures, while neural rankings vary across evaluation methods. Together, these findings show that behavioral and neural alignment are distinct targets: a constraint that improves the prediction of human behavior need not improve the prediction of brain activity.

1 INTRODUCTION

Do humans and Large Language Models (LLMs) process language in a similar way? David Marr's tri-level framework provides a principled lens for this question by decomposing information-processing systems into three levels (Figure 1A): the computational level (what is the goal of the computation?), the algorithmic level (what representations and step-by-step procedures are used?), and the implementational level (how are these steps physically realized?) (Marr, 1982). At the computational level, LLMs are primarily optimized for next-word prediction over massive text corpora. While predictive processing plays a central role in human language comprehension (Hale, 2001; Levy, 2008; Smith & Levy, 2013; Shain et al., 2024), next-token prediction is at best a rough proxy: biological language processing is also driven by communicative intent, pragmatic inference, multi-step predictive horizons, and social or goal-directed rewards (Bender & Koller, 2020;

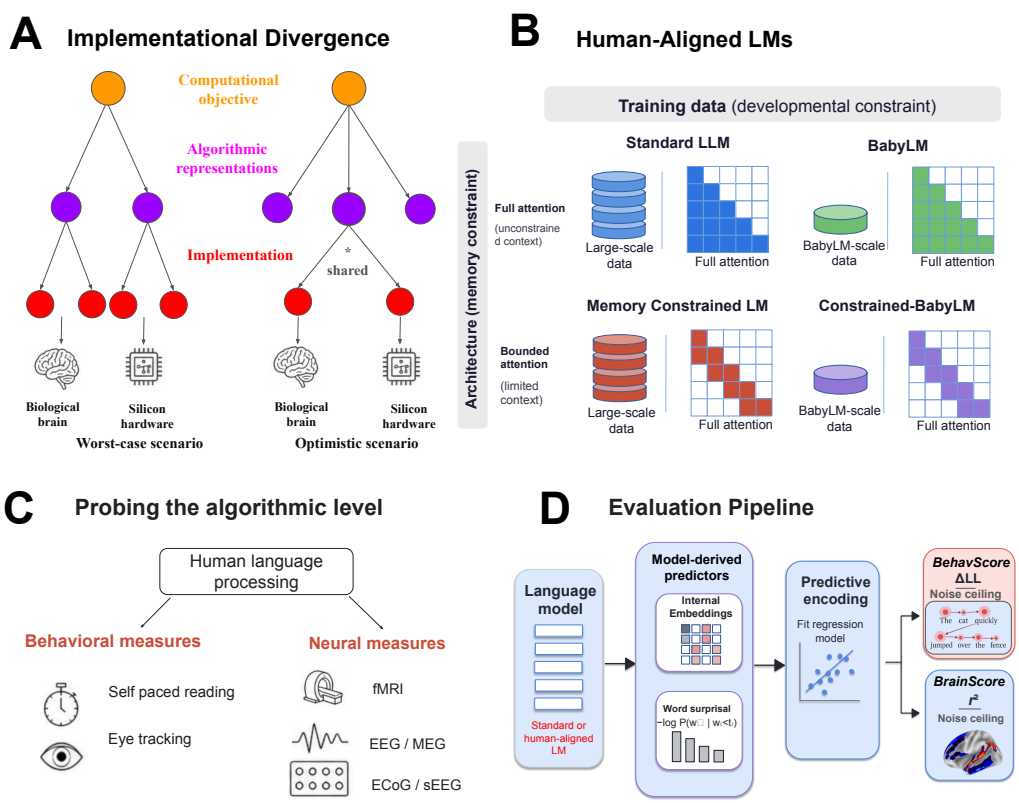


Figure 1: Overview of the study. (A) Marr's tri-level framework applied to humans and LLMs: even where the two systems share the same algorithmic-level representations (optimistic scenario), they may still diverge at the computational and implementational levels. (B) Human-aligned language models introduce either architectural constraints (limited attention, memory decay) or data constraints (developmentally limited training exposure) relative to standard LLMs. (C) Human language processing is probed through behavioral measures (self-paced reading, eye-tracking) and neural measures (fMRI, EEG/MEG, ECoG). (D) Final-layer LLM predictors, internal representations and word surprisal, are used to predict held-out behavioral and neural data (BehavScore and BrainScore), and are complemented by representational comparisons. Neither kind of correspondence alone establishes a shared algorithm.

Andreas, 2022). At the implementational level, the differences are clearer still. Standard LLMs rely on global self-attention over long context windows (Vaswani et al., 2017), uniform stacks of Transformer blocks that lack the modular organization of the cortical language network, non-local backpropagation rather than local learning rules, and training exposure spanning trillions of tokens, orders of magnitude beyond a human lifetime (Gilkerson et al., 2017; Warstadt et al., 2023a).

Given these differences at the computational and implementational levels, do LLMs and humans converge on similar representations and processing steps at the algorithmic level? At one extreme, the mismatches drive LLMs and human brains to adopt fundamentally different procedures. At the other, aligned with the Platonic Representation Hypothesis (Huh et al., 2024), shared environmental pressures and task demands push biological and artificial systems toward the same representations despite their structural differences. Which of these outcomes holds, and for which aspects of language processing, remains an open empirical question.

Cognitive scientists and neuroscientists have responded by building models that incorporate explicit human constraints, with the aim of moving them closer to human processing at the algorithmic level. On the architectural side, models introduce human cognitive limitations such as working-memory bottlenecks (Lewis & Vasishth, 2005), lossy or decaying context (Futrell et al., 2020),

recurrent processing dynamics (Michaelov et al., 2024a), or localized attention (de Varda & Marelli, 2024; Kuribayashi et al., 2022). On the data side, initiatives such as the BabyLM Challenge restrict training to developmentally realistic, child-scale input of 10 to 100 million words rather than web-scale text (Figure 1B; Warstadt et al., 2023a;b). These choices change the information available to a model and the representations it can learn. Whether they produce closer correspondence with human data, or whether standard LLMs already achieve comparable correspondence through scale, is an empirical question.

In humans, the algorithmic steps of language processing are probed through behavioral and neural measures (Figure 1C). Behavioral measures such as self-paced reading times and eye-tracking fixations index moment-by-moment processing difficulty (Futrell et al., 2021; Frank et al., 2013; Luke & Christianson, 2018; Siegelman et al., 2022). Neural measures capture complementary spatio-temporal scales: fMRI provides whole-brain spatial coverage (Schrimpf et al., 2021); EEG tracks temporal dynamics such as the N400 (Michaelov et al., 2024b; Federmeier et al., 2007); MEG and intracranial recordings combine finer temporal resolution with, for the latter, direct cortical access. A model that captures variation in reading difficulty need not be the model that most closely matches neural responses. This distinction matters when evaluating human-inspired models: a constraint intended to approximate human memory or training exposure may improve one aspect of alignment without improving another. Rather than interpreting model internals in isolation, we ask how well model-derived features, word surprisal and contextual embeddings, predict human behavioral and neural responses, and how closely the representational geometry of the model matches that of the responses (Figure 1D).

We test these questions in a 2×2 design that crosses an architectural memory constraint with a developmental data constraint: (1) memory-constrained architectures trained at web scale, (2) the same architectures trained on BabyLM-scale data, (3) standard architectures trained on BabyLM-scale data, and (4) standard, off-the-shelf LLMs with neither constraint. We evaluate 118 frozen models on 18 datasets spanning self-paced reading, eye-tracking, fMRI, EEG and N400 responses, MEG, and intracranial recordings, using final-layer embeddings and surprisal as predictors. We first identify individual models that offer competitive combinations of behavioral and neural prediction, using Pareto fronts. Because the four groups differ by orders of magnitude in parameter count and language-modeling quality, we then estimate group associations separately for behavioral and neural outcomes, adjusting for parameter count and perplexity. Finally, we compare direct prediction with unfitted representational similarity analysis (RSA) and confound-controlled regression on representational dissimilarity matrices (RDMs).

The central result is a behavioral–neural distinction. For contextual embeddings, data-constrained models predict reading behavior better than size- and perplexity-matched standard LLMs, but not neural responses, while memory-constrained models predict neural responses worse; both domain differences are tested directly and survive modeling the dependence between related models, a lineage cluster bootstrap, and dataset exclusions. Combining the two constraints cancels the behavioral advantage. Surprisal-based constraint effects are weaker and depend on datasets with unstable noise ceilings. Size also has different associations across outcomes: larger models have higher neural embedding scores, whereas behavioral surprisal prediction declines with size conditional on perplexity. Representational evaluations reinforce the competitiveness of small and constrained models for behavior, while revealing that neural rankings depend on the scoring method.

We use *alignment* to denote these measured correspondences. Correspondence in predictions or representational geometry can inform models of cognition without identifying a shared algorithm (Marr, 1982). Our contribution is therefore a comparative account of where constraints are associated with closer correspondence to human data, rather than a single verdict on whether constrained or standard language models are more human-like. Specifically:

- We evaluate 118 language models from four constraint groups on 18 behavioral and neural datasets (27 dataset units) with both embedding and surprisal predictors, under one common pipeline, and release it as the HumanScore-Language benchmark.
- We separate the association of each human-inspired constraint from that of model size and perplexity, with separate analyses for behavioral and neural responses and an explicit test of the interaction between the two constraints.

- We show, with inference that accounts for the dependence between related models and datasets and directly tests the difference between domains, that a developmental data budget is associated with better behavioral but not neural prediction and a memory constraint with worse neural prediction, and that their combination is not additive.
- We show that model rankings differ between direct prediction, unfitted RSA, and confound-controlled RDM regression, so that no single evaluation suffices to rank models by human-likeness.

2 Related Work

Predicting behavioral and neural responses. A growing body of work asks whether LLM representations line up with human brain activity and behavior during language processing. Model-derived features, most often next-word surprisal or hidden-state embeddings, predict fMRI and MEG responses during naturalistic listening and reading (Schrimpf et al., 2021; Caucheteux & King, 2022; Caucheteux et al., 2023), explain variance in fMRI and ECoG recordings beyond simpler linguistic features (Toneva & Wehbe, 2019), and track a hierarchy of predictive signals across the cortical language network (Caucheteux et al., 2023). These correspondences hold for models trained on child-scale corpora (Hosseini et al., 2024), can be strengthened by instruction tuning (Aw et al., 2024), and are matched or exceeded by recurrent architectures on several comprehension metrics (Michaelov et al., 2024a). Alignment does not grow indefinitely with training, however. Tracking eight Pythia models (14M to 6.7B parameters) across 34 training checkpoints on five neural benchmarks, AlKhamissi et al. (2025) find that alignment with the human language network rises sharply early in training, peaks after roughly 8B tokens and then saturates, following formal linguistic competence (knowledge of linguistic rules) more closely than functional competence, which keeps improving. They also report that, once controlling for the number of features, model size is not a reliable predictor of brain alignment, and that the correlation of brain alignment with next-word prediction and with behavioral alignment fades once models surpass human language proficiency. Intermediate layers already encode next-token distributions in a way that resembles human online processing (Kuribayashi et al., 2025), and reading times can be decoded from hidden states of models never trained to predict them (Tsipidi et al., 2026). Taken together, these findings motivate the Platonic Representation Hypothesis (Huh et al., 2024): models trained with different objectives, architectures, and data settle into overlapping representational geometry because they solve related tasks over similar statistics.

Surprisal and reading behavior. Surprisal links language-model predictions to processing difficulty (Hale, 2001; Levy, 2008; Smith & Levy, 2013) and remains the dominant predictor of reading times (Goodkind & Bicknell, 2018; Wilcox et al., 2020; 2023), with psychometric predictive power now benchmarked across model families and sizes (Kuribayashi et al., 2024). Its relationship with language-model quality is not monotonic: several studies report that smaller or less accurate models fit reading times better than large ones (Oh & Schuler, 2023b;a; Kuribayashi et al., 2021; Wilcox et al., 2024). Related work has generalized what counts as an anticipation signal in online processing (Giulianelli et al., 2024; Opedal et al., 2024). Neural encoding studies, by contrast, typically find that larger models predict brain responses better (Schrimpf et al., 2021; Caucheteux & King, 2022). These evaluations need not favor the same model, which is why our design keeps behavioral and neural outcomes separate and evaluates both embeddings and surprisal within each domain.

Memory and developmental constraints. If shared task pressure suffices to produce convergence, why constrain a model at all? One answer, pursued mostly in psycholinguistics, is that human sentence processing operates under resource limits that standard LLMs do not share, and that building those limits in should make a model behave more like a human reader. On the architecture side, this has taken the form of restricting attention itself: single-head or otherwise capacity-limited attention (Timkey & Linzen, 2023), distance-based decay that biases attention toward nearby tokens (Press et al., 2021; de Varda & Marelli, 2024; Clark et al., 2025; Thamma & Heilbron, 2025), content-sensitive forget gates (Lin et al., 2025) or stick-breaking attention (Tan et al., 2025), and memory-bounded or resource-rational formulations of the encoding process (Hahn et al., 2022; Lieder & Griffiths, 2019; Xu et al., 2026; Xu & Futrell, 2026; Rasmussen & Schuler, 2018; Keshev et al., 2025). These choices are usually motivated by capacity-limited theories of sentence processing, such as activation-based memory retrieval (Lewis & Vasishth, 2005) and lossy-context

surprisal (Futrell et al., 2020). On the data side, the BabyLM Challenge leaves the architecture alone but restricts training exposure to the 10–100 million words a child hears by early adolescence (Warstadt et al., 2023a;b; Hu et al., 2024), building on earlier work on pretraining with child-directed language (Huebner et al., 2021). We organize these approaches as two factors and explicitly estimate their interaction. Because the evaluated models also differ in architecture and training details, this grouping supports comparative associations rather than an isolated causal test of either constraint.

Prediction and representational correspondence. Encoding models assess information recoverable through a learned mapping. RSA instead compares relations among stimuli in model and response spaces (Kriegeskorte et al., 2008). These are distinct tests: a representation may support accurate decoding without matching response geometry under a specified distance metric, and a learned mapping has a capacity that grows with the width of the representation. Feature scaling also affects embedding distances (Timkey & van Schijndel, 2021). We therefore report both unfitted RSA and held-out RDM regression beyond lexical confounds, using item-wise partitions to separate training and test stimuli (Schütt et al., 2023). This combination lets us assess which model comparisons recur across evaluations and which are specific to a learned response mapping or a representational score.

Closest work. Concurrently with our work, de Varda et al. (2026) compare a low-dimensional effort predictor (surprisal, frequency, and word length) with high-dimensional final-layer embeddings across eight behavioral and five neural datasets, several of which overlap with ours (Natural Stories, Provo, MECO, ZuCo, Tuckute, UCL). They report that behavioral measures are well explained by the effort predictor alone, whereas brain responses require the richer embedding space, and that this split coincides with opposite scaling trends within the GPT-2 family: smaller models fit behavior better, larger models fit the brain better. Their question is largely orthogonal to ours. They hold the model class fixed and ask which *predictor* best explains each modality; we hold the predictors fixed and ask which *model class*, architecturally or developmentally constrained versus standard, best explains each modality, across a much larger sweep of models and datasets. Our size coefficients are consistent with their behavioral–neural scaling contrast for surprisal, and extend it to the question of human-inspired constraints.

Our study thus differs from prior work in two ways. First, rather than taking a side in the constrained-versus-standard debate, we place memory-constrained architectures, BabyLM-trained models, and standard LLMs in the same evaluation pipeline, so that any advantage for the human-inspired models appears as a direct comparison rather than being read across papers with different datasets and metrics. Second, we evaluate this comparison across self-paced reading and eye-tracking corpora together with fMRI, EEG/ERP, MEG, and intracranial recordings, and with both predictive and representational measures, so that conclusions about alignment are not artifacts of any single modality, measure, or evaluation method.

3 Experimental Design

3.1 Question and design

We ask whether building a human cognitive or developmental constraint into a language model is associated with closer correspondence to human behavioral and neural data than a standard, unconstrained LLM provides. We operationalize correspondence in two ways: predictive power, i.e. how well features derived from a frozen model predict held-out human responses, and representational similarity between model and response dissimilarity matrices. Neither is a test of shared mechanism.

Models are organized in a 2×2 design (Section 3.2): a memory factor (M; a memory-limited architecture or not) crossed with a data factor (D; a developmental, BabyLM-scale training corpus or web-scale text). This gives three questions: (H1) is a memory limitation associated with a change in alignment, (H2) is a developmental data budget, and (H3) do the two combine additively or interact. Because the four groups differ greatly in parameter count, depth and language-modeling quality (Table 7), a raw comparison of group means would confound the constraint with scale. We therefore compare groups conditional on size and perplexity (Section 3.5), and answer each question separately for behavioral and neural responses.

Every dataset is crossed with every model and with two feature types (Section 3.4), giving a 2×2 of {neural, behavioral} data $\times$ {embedding, surprisal} predictors. The evaluation procedure is identical across models; only per-dataset configurations differ.

3.2 MODELS

We evaluate 118 frozen models, organized into four groups (Table 7, Appendix B). Standard LLMs have $M = D = 0$; Memory-Constrained and BabyLM models have $(M, D) = (1, 0)$ and $(0, 1)$; Constrained-BabyLM models have $M = D = 1$. No model is fine-tuned on any behavioral or neural target.

Standard LLMs (59 models). Off-the-shelf, web-scale autoregressive models with neither constraint: GPT-2, GPT-Neo, OPT, Pythia, OLMo-2, Qwen3, Llama-3, Ministral, and the recurrent and state-space models Mamba, Mamba2 and RWKV-4 (14M to 66B parameters, 6 to 65 layers), plus the two matched baselines released with the Forgetting Transformer (FoX), which have ordinary attention and no forget gate (760M parameters). Mamba, Mamba2 and RWKV-4 carry a fixed-size recurrent state but no constraint motivated by human memory; recurrent state alone does not define the tested memory-constraint category, so they are grouped here.

Memory-Constrained (24 models). Architectures that limit how much of the preceding context the model can access or weight, motivated by capacity-limited accounts of human sentence processing such as activation-based retrieval (Lewis & Vasishth, 2005) and lossy-context surprisal (Futrell et al., 2020): distance-penalized attention (ALiBi), forget-gate and stick-breaking replacements for softmax attention, hard n-gram windows, a small bounded-state recurrent network (CBR-RNN), and web-scale models with the same mechanisms (BLOOM, whose attention uses ALiBi; FoX; MoLM). These are trained on web-scale text (15M to 7.1B parameters, 1 to 49 layers).

BabyLM (11 models). Standard Transformer architectures trained on roughly the number of words a child hears by early adolescence, 10M to 100M words, following the BabyLM Challenge (Warstadt et al., 2023a;b; Hu et al., 2024) (33M to 140M parameters, 11 to 13 layers).

Constrained-BabyLM (24 models). The ALiBi, forgetting, stick-breaking and hard n-gram architectures trained on the BabyLM corpora, i.e. both constraints at once (15M to 48M parameters, 2 to 4 layers).

Keeping the last group separate makes the crossing explicit and lets the regression ask whether the two constraints add up or interact. The categorization concerns the selected architectural mechanisms and training budgets, not a claim that all models within a group share one cognitive mechanism. Pythia and OLMo-2 intermediate training checkpoints are available but are not part of the comparison.

3.3 HUMAN DATA

Every dataset is reduced to the same unit of analysis, a word (or sentence, or item) embedded in its preceding context and paired with one or more measured human responses, so any model can be run through any dataset without dataset-specific model code. In total we use 18 datasets, which we split into 27 *dataset units*: reading-time measures that come from the same recording (first-pass, go-past and scan-path times) and the three ZuCo reading tasks are counted as separate units, giving 16 behavioral and 11 neural units (Table 8, Appendix C). Units of one corpus share their items and participants and are therefore not independent; we return to this in Section 3.5.

Behavioral datasets. Behavioral measures index processing difficulty indirectly, through the time a reader takes to move past a word.

- **Self-paced reading.** Natural Stories (Futrell et al., 2021), naturalistic narratives deliberately written to contain rare syntactic constructions, and UCL Reading (Frank et al., 2013), 361 sentences read by 117 participants.

- **Eye-tracking.** GECO (Cop et al., 2017), Provo (Luke & Christianson, 2018), the English part of MECO (Siegelman et al., 2022), OneStop (Berzak et al., 2025), and the eye-tracking part of ZuCo (Hollenstein et al., 2018), where participants read isolated sentences under three task instructions (normal reading, sentiment judgment, task-specific relation judgment). Depending on the corpus we use first-pass duration, go-past time and scan-path (total dwell) time, each as its own unit.

Neural datasets. Neural measures record language processing from the brain at complementary spatial and temporal resolutions.

- **fMRI.** Le Petit Prince (Li et al., 2022) (audiobook listening), Natural Stories fMRI (Shain et al., 2022) (responses pre-averaged within functionally localized language and multiple-demand regions), Tuckute et al.'s language-network dataset (Tuckute et al., 2024) (isolated sentences), and Pereira et al. (Pereira et al., 2018) (isolated sentences from two experiments).
- **EEG / ERP.** Broderick et al.'s continuous-listening audiobook corpus (Broderick et al., 2018), and three N400 datasets built around cloze-probability manipulations: a nine-site replication of the classic article/noun expectancy design (Nieuwland et al., 2018), and two studies from the Szewczyk and Federmeier collection (Szewczyk & Federmeier, 2022), Federmeier et al. (2007) (Federmeier et al., 2007) and Hubbard et al. (2019), analyzed separately.
- **Intracranial recordings.** A sentence-reading intracranial dataset (Fedorenko et al., 2016) and a large ECoG dataset recorded during a naturalistic podcast (Zada et al., 2025).
- **MEG.** MEG-MASC (Gwilliams et al., 2022), passive listening to naturalistic stories.

Each unit retains its native word, sentence, item, or time-aligned response format.

3.4 Final-layer predictors and direct prediction

For each model and dataset we compute two kinds of predictors from the same forward pass over the full preceding context. All reported analyses use the final hidden layer or the final output distribution. Models have between 1 and 65 layers, so "the representation of a model" needs a rule that is the same for all of them; the final layer involves no selection and treats shallow and deep models alike.

Embedding track. We extract the final-layer hidden state at every word, mean-pooled over its subword tokens (sentence datasets use pooled sentence features, and continuous recordings require temporal alignment, Appendix D). A ridge regression maps these embeddings to the human response, with the regularization strength chosen by nested cross-validation and all scores computed on held-out folds (stories, passages, sentences or time chunks, depending on the dataset; Table 9). The score is the held-out Pearson correlation between predicted and observed responses. The embedding track uses the full representation, so it measures how much of the signal is linearly decodable from the model. The ridge encoders include no covariates: unlike the surprisal track, they are not given word length or frequency, so lexical information that a model's representation carries can contribute to the score.

Surprisal track. Surprisal is the model's next-token negative log probability, summed over the subword tokens of a word. For each dataset we fit a linear mixed-effects model with dataset-specific nuisance predictors (word length and frequency for reading data, plus the covariates required by each neural recording design) and compare it with the model that adds the current word's surprisal and, where specified, the previous word's surprisal as a spillover term. Random-effect structures are dataset-specific (Appendix G). The score is the improvement in held-out log likelihood per test observation,

$$\Delta\ell = \frac{\ell^{\text{test}}_{\text{full}} - \ell^{\text{test}}_{\text{base}}}{n_{\text{test}}}. \tag{1}$$

A model whose surprisal adds nothing beyond the nuisance predictors scores 0, regardless of its size; negative values indicate worse held-out fit than the baseline. The observation is defined by the dataset and need not be a single word in the neural analyses. The two predictor tracks consequently measure different properties and use different outcome scales.

3.5 COMPOSITE SCORES AND GROUP ANALYSIS

Normalization, BrainScore and BehavScore. Raw scores (correlations, $\Delta\ell$) are not comparable across datasets: a correlation of 0.3 may be close to the achievable maximum for a noisy single-trial recording and far from it for a heavily averaged fMRI response. For embeddings, each dataset-unit score is therefore divided by its split-half correlation noise ceiling (van Bree et al., 2025); for surprisal, it is divided by a split-half $\Delta\ell$ reference built from the human data (Timkey et al., 2026) (Appendix A). We call the median normalized score across neural units a model's *BrainScore* and the median across behavioral units its *BehavScore*; each is defined for both predictor tracks. These ratios are not fractions of explained variance, and split-half likelihood references can be exceeded (Appendix A).

Pareto fronts. A model is Pareto-optimal if no other evaluated model scores at least as highly on both BrainScore and BehavScore and strictly higher on one. Front membership describes estimated scores without uncertainty-based selection.

Separate behavioral and neural regressions. For each predictor track we regress the normalized score of each (model, dataset unit) pair on the two constraint indicators, the size of the model, and its language-modeling quality on that dataset's text, separately within each response domain $m \in \{\text{behavioral}, \text{neural}\}$:

$$\begin{aligned} Y_{ij}^{(m)} = \beta_0^{(m)} + \beta_M^{(m)} M_i + \beta_D^{(m)} D_i + \beta_{MD}^{(m)} M_i D_i + \beta_S^{(m)} \log_{10} P_i + \beta_Q^{(m)} \log_{10} \mathrm{PPL}_{ij} \\ + u_j + b_{Sj} \log_{10} P_i + b_{Qj} \log_{10} \mathrm{PPL}_{ij} + f_{\ell(i)} + v_i + c_{k(j)} + \epsilon_{ij}. \end{aligned} \tag{2}$$

Here Y is the noise-ceiling-normalized direct-prediction score of model i on unit j, P is parameter count, and PPL is perplexity on the dataset text (both centered). Standard LLMs are the reference. β_M and β_D are the contrasts of each single-constraint group with Standard LLMs, the combined contrast of Constrained-BabyLM models is $\beta_M + \beta_D + \beta_{MD}$, and β_S and β_Q absorb scale: a constraint is associated with an advantage only if its contrast is positive *beyond* what size and perplexity predict. An interaction measures departure from additivity, not the combined group's advantage over the reference.

The random effects model the dependence in the data. Scores of one dataset unit share a random intercept u_j and unit-specific slopes b_{Sj} and b_{Qj} for size and perplexity, because the association of size with alignment varies in strength and sign across datasets (Appendix H.5). Models of one lineage share a random intercept $f_{\ell(i)}$: a lineage is a model family (e.g. all ALiBi models, whatever their training data, or all Pythia models), except that the eleven BabyLM models, which come from five independent submissions, are split into those submissions; there are 25 lineages. Each model has its own random intercept v_i, since the same model contributes one score per dataset unit, and behavioral units recorded in one corpus (first-pass, go-past and scan-path times) share a corpus intercept $c_{k(j)}$. Models are fit by restricted maximum likelihood with `lme4`, and tests use Satterthwaite degrees of freedom. The analysis includes the 115 models with usable scores and perplexities (56 Standard LLMs, 24 Memory-Constrained, 11 BabyLM, and 24 Constrained-BabyLM; the three OLMo-2 models are excluded because their perplexities were not computed), giving 1,822 behavioral and 1,257 neural rows in each track.

Testing the behavioral–neural difference. Separate fits show whether a constraint is associated with each domain, but a significant coefficient in one domain and a nonsignificant one in the other is not evidence that they differ. We therefore also fit a joint model of both domains, in which every fixed effect of Equation 2 interacts with domain, with the same random effects plus lineage-by-domain and model-by-domain intercepts. The domain interactions directly estimate the difference between neural and behavioral coefficients.

Robustness. We assess how much the conclusions depend on the analysis choices in five ways (Appendix E): (i) a specification with only a dataset-unit random intercept, which treats the rows as otherwise independent; (ii) a cluster bootstrap that resamples model lineages with replacement within each constraint group; (iii) refits that exclude the dataset units whose likelihood references are flagged as unstable (Appendix A.3), winsorize scores within unit, exclude the GPT-BERT models, or add model depth as a covariate; (iv) leave-one-unit-out refits; and (v) the same models on a within-unit robust z-score, $(x - \text{median})/(\text{IQR}/1.349)$ over models, which expresses effects in units of

the between-model spread and does not depend on the noise ceilings. We base our conclusions on effects that hold across these checks. The groups still differ in architecture, objective and training corpus beyond the two binary factors, so the estimates are adjusted associations, not causal effects.

3.6 Representational evaluation

Ridge encoding models have a number of free parameters that grows with the hidden size of the model, so a score that increases with model size may partly reflect the capacity of the regression. We therefore complement direct prediction with two analyses of representational dissimilarity matrices (RDMs; Kriegeskorte et al., 2008), in which a model contributes a single stimulus-by-stimulus matrix whatever its width. Embedding distances are one minus the Pearson correlation between final-layer vectors after standardizing each feature across stimuli (Timkey & van Schijndel, 2021). Surprisal distances are absolute differences in surprisal. Response distances are correlation distances for neural patterns and absolute differences for scalar reading-time or N400 responses.

Unfitted Pearson RSA correlates corresponding model and response distances without learning a response mapping. RDM regression instead measures the held-out Gaussian log-likelihood improvement when a model RDM is added to word-length and frequency RDMs, mirroring Equation 1. Training and test partitions separate stimuli rather than randomly separating pairs (Schütt et al., 2023). Both analyses yield ceiling-normalized behavioral and neural composites, but their numerical scales differ from direct prediction. The representational fronts cover all 118 models, 16 behavioral measures, and eight neural datasets; Le Petit Prince, Natural Stories fMRI, and Fedorenko are absent. Appendix H specifies the scoring and aggregation.

4 Results

Our analysis proceeds in steps, each motivated by the limits of the previous one. We first ask which individual models offer the best combination of behavioral and neural prediction, using Pareto fronts. Because these fronts compare models that differ in size and language-modeling quality, we then estimate the association of each constraint with behavioral and neural scores at matched size and perplexity, and use matched model comparisons to locate where these associations come from. Finally, because direct prediction relies on a learned mapping whose capacity grows with model width, we ask whether the same picture holds under representational evaluations that fit no such mapping. Together, these analyses show that, for embeddings, a developmental data budget is associated with better behavioral prediction and a memory constraint with worse neural prediction, that the two constraints do not combine additively, and that neural model rankings depend on the evaluation method.

4.1 Different models lead behavioral and neural prediction

To obtain a first overview of how the 118 models trade off behavioral and neural alignment, we plot each model's BehavScore against its BrainScore and identify the Pareto front, i.e. the models that no other model outperforms on both axes at once. No evaluated model maximizes both behavioral and neural scores in either feature track (Figure 2). Standard LLMs occupy most frontier positions, alongside specific constrained models. The embedding frontier contains five Standard LLMs, BLM-GPT2s-90M from BabyLM, and the memory-constrained BLOOM-3B. Its behavioral endpoint is OPT-350M, while OPT-66B has the highest neural composite. The surprisal frontier contains four Standard LLMs and BLOOM-1.1B: BLOOM-1.1B leads behavior, whereas Ministral-3-14B leads the neural composite. Neither direct-prediction frontier contains a Constrained-BabyLM model. The models that lead behavioral prediction are thus different from, and typically much smaller than, those that lead neural prediction, a first indication that the two domains reward different model properties.

These fronts identify competitive individual models, but they do not tell us whether a constraint is associated with better or worse alignment as a group. To address this question, we first compare group medians directly.

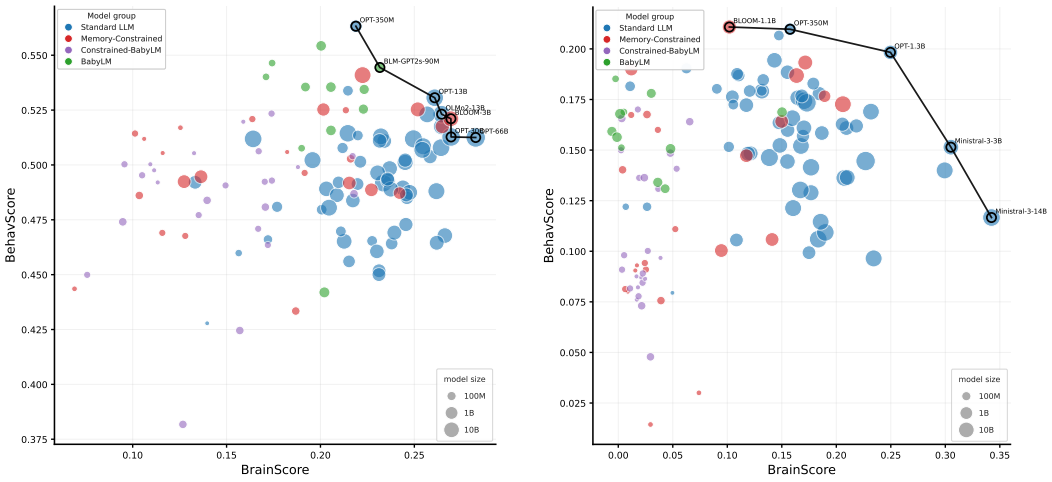


Figure 2: Direct-prediction Pareto fronts for final-layer embeddings (left) and surprisal (right). Axes are medians of ceiling-normalized dataset-unit scores. Colors identify model groups and marker sizes encode parameter count on a logarithmic scale. Black outlines identify nondominated models. The two tracks have different score definitions; their axis values are not directly comparable.

Table 1: Unadjusted group medians of the composite scores (median across models within a group of each model's BehavScore or BrainScore; last layer, ceiling-normalized). N is the number of models with at least one finite score. These raw differences confound the constraints with size and perplexity; Table 2 adjusts for both.

		Embeddings		Surprisal	
Group	N	BehavScore	BrainScore	BehavScore	BrainScore
Standard LLMs	59	0.492	0.232	0.161	0.143
Memory-Constrained	24	0.504	0.185	0.126	0.055
BabyLM	11	0.535	0.202	0.159	0.030
Constrained-BabyLM	24	0.492	0.153	0.094	0.029

4.2 Unadjusted differences between groups

Table 1 gives the median composite score of each group, and it already shows the behavioral–neural asymmetry suggested by the fronts. Standard LLMs have the highest median BrainScore in both tracks, and the gap is large for surprisal (0.143, against 0.055 for Memory-Constrained and about 0.03 for both BabyLM groups). For behavior the ordering differs: with embeddings, BabyLM models have the highest median BehavScore (0.535, against 0.492 for Standard LLMs), and with surprisal, BabyLM and Standard LLMs are close (0.159 and 0.161). Constrained-BabyLM models have the lowest or tied-lowest median in every column. These raw differences do not isolate the constraints, because the groups differ greatly in size, depth and language-modeling quality (Table 7). A constrained model could score lower than a Standard LLM simply because it is smaller or a worse language model. To separate the constraints from these factors, we performed a matched-scale analysis: the regressions of Equation 2, fit separately for behavioral and neural scores, which compare each group with Standard LLMs of the same size and perplexity.

4.3 For embeddings, constraints are associated with behavioral and neural prediction in opposite directions

Table 2 reports the regressions of Equation 2. A positive constraint coefficient means that, relative to a Standard LLM of the same size and perplexity, models with that constraint have higher normalized scores in that response domain. Coefficients are in normalized-score units, so their magnitudes are comparable within a column but not across tracks or domains, whose score ranges differ (Ap-

Table 2: Group associations in final-layer direct prediction (Equation 2, preferred specification). Coefficients are in noise-ceiling-normalized score units with standard errors in parentheses (Satterthwaite degrees of freedom). Behavioral and neural columns are separate fits with random intercepts for dataset unit, model lineage, model and corpus, and dataset-unit random slopes for size and perplexity. The Neural − behavioral columns come from a joint model of both domains with domain interactions and directly test whether a coefficient differs between them. Single-constraint and combined contrasts are relative to Standard LLMs conditional on parameter count and perplexity. $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$. Robustness checks are in Table 10.

	Embeddings			Surprisal		
	Behavioral	Neural	Neural − behav.	Behavioral	Neural	Neural − behav.
Memory only	.013 (.008)	$-.031$ (.010)**	$-.047$ (.010)***	$-.001$ (.008)	$-.092$ (.065)	$-.089$ (.060)
Data only	.033 (.012)**	$-.006$ (.013)	$-.042$ (.014)**	.022 (.011)	$-.367$ (.084)***	$-.391$ (.079)***
Memory × data	$-.040$ (.012)**	.022 (.014)	.064 (.015)***	$-.026$ (.012)*	.384 (.092)***	.417 (.086)***
Both constraints	.006 (.009)	$-.016$ (.011)	$-.025$ (.011)*	$-.005$ (.009)	$-.075$ (.072)	$-.063$ (.065)
$\log_{10}$ parameters	.004 (.003)	.026 (.012)	.024 (.010)*	$-.035$ (.014)*	.283 (.192)	.316 (.159)
$\log_{10}$ perplexity	$-.005$ (.003)	$-.020$ (.009)*	$-.009$ (.008)	$-.058$ (.024)*	$-.105$ (.111)	$-.049$ (.095)
Rows / dataset units	1,822 / 16	1,257 / 11	joint	1,822 / 16	1,257 / 11	joint

pendix A.3). Because models of one lineage, scores of one model, and units of one corpus are not independent, the random effects of Equation 2 widen the standard errors considerably relative to a fit with a dataset-unit intercept only (Appendix E); we therefore distinguish associations that are robust to the checks described in Section 3.5 from those that are not.

We begin with the data constraint in the embedding track, which gives the clearest result. Conditional on size and perplexity, BabyLM models without a memory constraint have higher behavioral embedding scores than Standard LLMs ($\beta_D = .033$, SE .012, $p = .008$). This advantage holds in every robustness check, including the lineage bootstrap, the exclusion of the GPT-BERT models, the addition of depth as a covariate, and every leave-one-unit-out refit (Table 10). On the robust z scale it corresponds to about one between-model standard deviation (1.05, 95% CI $[0.30, 1.81]$), a large difference relative to how much models differ from each other. The corresponding neural contrast is close to zero ($-.006$, SE .013), and the joint model confirms that the data contrast differs between the two domains (neural − behavioral: $-.042$, SE .014, $p = .007$).

The memory constraint shows the mirror-image pattern. Memory-constrained models have lower neural embedding scores than Standard LLMs of the same size and perplexity ($\beta_M = -.031$, SE .010, $p = .002$), an association that also survives every robustness check. Their behavioral embedding contrast is positive but not reliably different from zero (.013, SE .008, $p = .12$); the advantage that a dataset-intercept-only fit suggests (.016, $p < .001$) does not survive modeling the dependence between models. The joint model again shows that the two domains differ ($-.047$, SE .010, $p < .001$). For embeddings, the two constraints are thus associated with prediction in opposite directions across the two domains: the data budget with better behavioral prediction, and the memory constraint with worse neural prediction.

Since each constraint alone is associated with an advantage or no difference for behavior, one might expect models combining both to benefit most. The interaction shows the opposite. The memory-by-data interaction for behavioral embeddings is negative ($\beta_{MD} = -.040$, SE .012, $p = .002$) and robust, and it cancels the data advantage: Constrained-BabyLM models do not differ from Standard LLMs in either domain (.006, SE .009, behavioral; $-.016$, SE .011, neural).

The surprisal track gives a less clear picture. The behavioral data contrast is positive but marginal ($\beta_D = .022$, SE .011, $p = .062$). It is reliable on the robust z scale ($p = .016$) but disappears when UCL self-paced reading, whose likelihood reference is exceeded by many models, is excluded (.008, $p = .33$). The neural surprisal contrasts are large in Table 2 (data only $-.367$, memory × data .384, both $p < .001$), but they are driven by the four units whose likelihood references are flagged as unstable: without them, the data contrast is $-.030$ ($p = .21$) and the interaction .039 ($p = .15$). We therefore do not draw conclusions from the neural surprisal constraint coefficients, and treat the surprisal track as inconclusive about the constraints.

Table 3: Design-based comparisons. Within architecture: the six memory-limited architectures of Bu et al. (2026), each trained on BabyLM 10M, BabyLM 100M and the Pile at 2 and 4 layers, so that architecture, code and size are fixed and only the training data differ (random intercepts for architecture, model and unit; no size or perplexity adjustment, since perplexity is an outcome of the data). Matched pairs: the two Forgetting Transformer models against their own softmax-attention baselines (mean difference across dataset units; p from a t test across units). By mechanism: the memory-only contrast of Table 2 estimated separately for each memory mechanism. Coefficients (SE) in normalized score units; $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$.

	Embeddings		Surprisal	
	Behavioral	Neural	Behavioral	Neural
Within architecture (36 models; reference: Pile)				
BabyLM 100M	$-.006$ (.005)	.027 (.010)*	$-.005$ (.008)	.052 (.034)
BabyLM 10M	$-.012$ (.005)*	.004 (.010)	$-.006$ (.008)	.038 (.034)
Matched pairs (2 pairs, same data and size)				
Forget gate − softmax baseline	$-.001$ (.004)	$-.039$ (.018)	$-.004$ (.005)	.006 (.094)
Memory only, by mechanism (vs Standard LLMs, preferred specification)				
Distance bias (ALiBi, BLOOM)	.008 (.014)	$-.026$ (.013)	.016 (.014)	$-.109$ (.093)
Forget gate (Forgetting, FoX)	.004 (.011)	$-.059$ (.012)***	$-.003$ (.011)	$-.153$ (.091)
Stick-breaking (incl. MoLM)	.012 (.014)	.009 (.013)	.013 (.014)	$-.100$ (.096)
Hard n-gram window	.041 (.014)**	$-.016$ (.014)	$-.030$ (.014)*	.044 (.104)
CBR-RNN	.037 (.025)	$-.025$ (.023)	.000 (.032)	.291 (.176)

4.4 Design-based comparisons locate the associations

The group contrasts compare lineages that differ in more than the two binary factors: the memory group pools several mechanisms, and the data contrast compares BabyLM submissions with web-scale LLMs of different architectures. Our model set contains three comparisons that hold more of these differences fixed (Table 3), and we used them to ask where the associations come from.

First, the six memory-limited architectures of Bu et al. (2026) were each trained on BabyLM 10M, BabyLM 100M and the Pile at two depths, so their training data vary while architecture, code and size stay fixed. Within these architectures, less training data does not improve behavioral prediction: models trained on 10M words predict reading behavior slightly worse from embeddings than the same architectures trained on the Pile ($-.012$, SE .005), and 100M-word models do not differ from them ($-.006$, SE .005). This agrees with the negative memory-by-data interaction of Table 2: for memory-limited architectures, a developmental data budget brings no behavioral benefit. The behavioral data advantage is therefore a contrast between BabyLM-trained standard architectures and web-scale LLMs, which we cannot reduce to the amount of training data alone; it does, however, persist without the six GPT-BERT models (Table 10), so it is not specific to one BabyLM submission. On the neural side, the 100M-word models predict neural responses from embeddings slightly better than the Pile models (.027, SE .010), but the 10M-word models do not, so we see no consistent neural dose-response either.

Second, the Forgetting Transformer was released with softmax-attention baselines of the same size, trained on the same data with the same code, giving two matched pairs that isolate the forget gate. The forget gate does not change behavioral prediction in either track, but lowers neural prediction from embeddings ($-.039$, SE .018, $p = .058$; lower in eight of eleven neural units). With only two pairs this comparison has little power, but it points in the same direction as the memory contrast of Table 2.

Third, estimating the memory contrast separately for each mechanism shows that it is not uniform. The neural embedding penalty comes mainly from forget-gate models ($-.059$, SE .012, $p < .001$) and, less clearly, from distance-biased attention ($-.026$, SE .014, $p = .055$), whereas stick-breaking attention and hard n-gram windows show no neural penalty. Conversely, hard n-gram windows are the only mechanism with a behavioral embedding advantage (.041, SE .014, $p = .006$), the same models that lead the behavioral end of the embedding RSA and RDM-regression fronts; for behavioral surprisal their contrast is negative ($-.030$, SE .014). The pooled memory contrast thus averages over mechanisms with different associations, and the neural memory penalty is best described as a property of forget-gate and distance-biased attention.

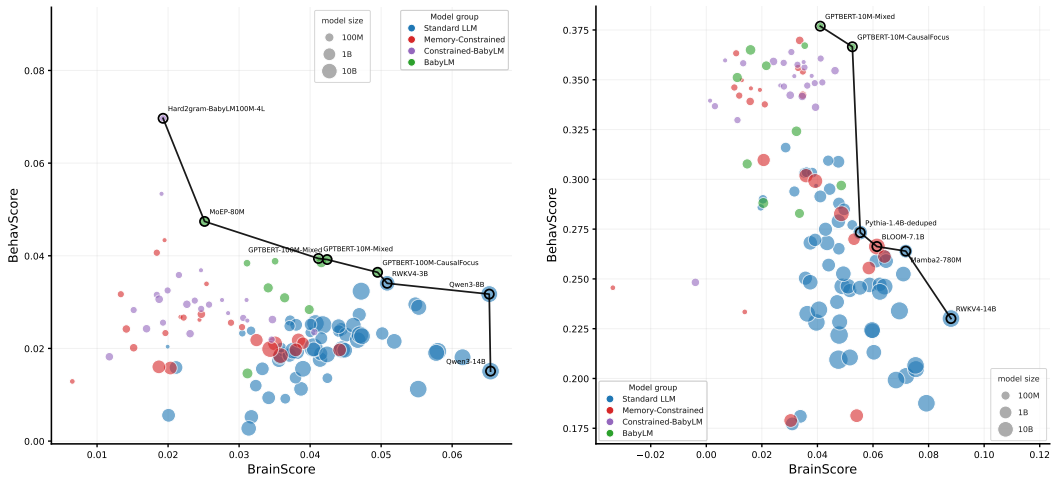


Figure 3: Unfitted Pearson RSA Pareto fronts: embeddings (left) and surprisal (right). Scores summarize model–response RDM correlations divided by their RSA ceilings over eight neural datasets and 16 behavioral measures. Small and constrained models occupy the behavioral end, while Standard LLMs attain the highest neural composites. These descriptive fronts assess representational geometry without fitting a response mapping.

4.5 Size and perplexity

The same regressions show how size and perplexity relate to alignment once group membership is accounted for. In the embedding track, parameter count is positively associated with neural prediction (.026 per tenfold increase, SE .012, $p = .052$ in the separate fit; .028, $p = .002$ in the joint model), and more strongly than with behavioral prediction (neural − behavioral .024, SE .010, $p = .025$). The size association for neural surprisal is positive but imprecise (.283, SE .192), reflecting large differences in slope across neural datasets.

For behavioral surprisal, both covariates have negative coefficients: lower perplexity is associated with better prediction ($\beta_Q = -.058$, SE .024, $p = .027$), and, at fixed perplexity, larger models predict reading behavior worse ($\beta_S = -.035$, SE .014, $p = .025$). Both associations hold in every robustness check, and they are stronger without the flagged UCL unit. Taken together, these findings do not support a universal increase in human alignment with model size: size is associated with better neural prediction from embeddings but, at matched language-modeling quality, with worse behavioral prediction from surprisal.

4.6 Representational comparisons support behavioral competitiveness and reveal evaluation-dependent neural rankings

The analyses so far rely on direct prediction, in which a learned ridge mapping (for embeddings) or a regression on surprisal relates model features to responses. The capacity of the ridge mapping grows with the width of the representation, which could favor large models for reasons unrelated to representational similarity. To test whether our conclusions depend on this learned mapping, we further performed an unfitted representational similarity analysis, which compares the geometry of model and response dissimilarity matrices directly and assigns every model a single, parameter-free score. The behavioral end of the embedding RSA frontier is occupied by Hard2gram-BabyLM100M-4L, followed by MoEP-80M and GPTBERT models; Qwen3-14B has the highest neural composite (Figure 3). For surprisal RSA, GPTBERT-10M-Mixed leads behavior, while RWKV4-14B leads neural alignment. Thus constrained models remain competitive for behavior under an evaluation that does not learn a response mapping. This agrees with the behavioral group findings of the matched-scale analysis, although it is a descriptive convergence rather than an independent adjusted test of the constraint coefficients.

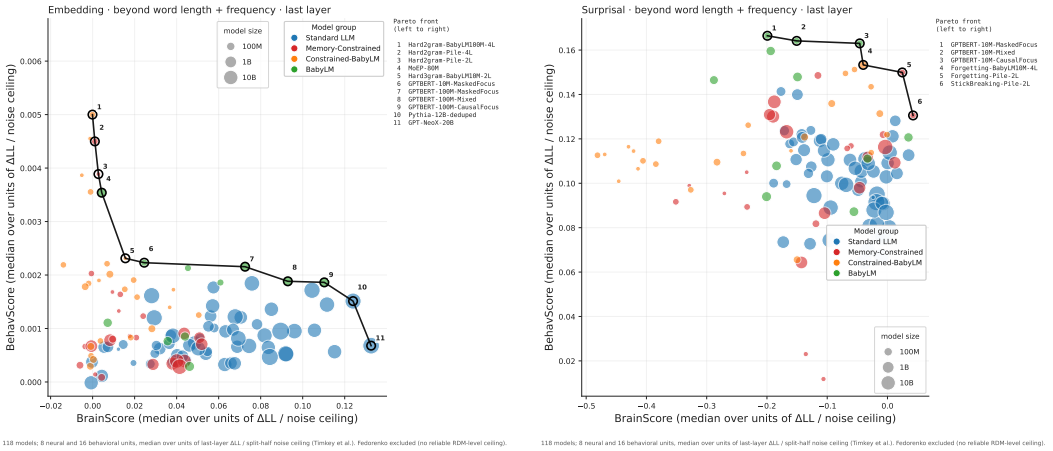


Figure 4: Confound-controlled RDM-regression Pareto fronts: embeddings (left) and surprisal (right). Scores are held-out $\Delta\ell$ per pair beyond word-length and frequency RDMs, divided by split-half references and aggregated across dataset units. Unlike unfitted surprisal RSA, the surprisal regression frontier includes no Standard LLM. Negative scores indicate poorer held-out fit than the confound-only baseline; most neural surprisal scores are negative.

Unfitted RSA, however, does not control for lexical properties: two words may have similar model representations and similar reading times simply because they have similar length or frequency. To rule out this explanation, we additionally fit an RDM regression, which measures how much a model RDM improves held-out prediction of the response RDM beyond word-length and frequency RDMs.

Controlling for these lexical distances preserves the behavioral competitiveness of constrained models but changes the neural rankings (Figure 4). Because the two axes differ greatly in how much a model RDM adds beyond the lexical baseline, we describe each in turn. For embeddings, the gain on the behavioral axis is small for every model (normalized BehavScores between 0 and .005), whereas most models improve neural prediction beyond the lexical RDMs (BrainScores from $-.014$ to .133, positive for 102 of 118 models). The embedding frontier includes all four groups: Hard2gram-BabyLM100M-4L leads behavior, followed by other hard-window and BabyLM models, and BabyLM models also have the highest median behavioral score of the four groups (.0019, against .0007 for Standard LLMs), while GPT-NeoX-20B leads the neural composite. For surprisal the situation is reversed. Behavioral gains are larger (BehavScores from .012 to .166), and the frontier consists entirely of BabyLM and constrained models: GPTBERT-10M-MaskedFocus leads behavior, and BabyLM models have the highest median behavioral score (.147, against .103 for Standard LLMs). On the neural axis, however, surprisal distances add little beyond the lexical RDMs: only 11 of 118 models have a positive BrainScore. The models at the neural end of the frontier, StickBreaking-Pile-2L (.042) and Forgetting-Pile-2L (.025), are memory-constrained models among these few, but Standard LLMs have the least negative median ($-.049$) and are absent from the frontier only because of their lower behavioral scores. The surprisal results therefore support the behavioral competitiveness of constrained models, but on the neural axis they show a ranking among models that mostly fail to improve on the lexical baseline, not good neural prediction.

Because the matched-scale analysis found opposite size associations for behavioral and neural prediction, we next asked whether representational similarity shows the same domain-dependent scaling. Across the 118 models, embedding RSA correlates positively with log parameter count in seven of eight neural datasets, but negatively in fourteen of sixteen behavioral measures (Table 13). MEG-MASC is the neural exception ($r = -.558$). Behavioral examples include MECO first-pass duration ($r = -.597$) and ZuCo task-specific reading ($r = -.643$). Because the smallest models are also the constrained ones, these pooled correlations could merely reflect group differences. Within constraint groups (partial correlations controlling for group), the pattern is weaker but keeps its direction: positive in seven of eight neural datasets and negative in twelve of sixteen behavioral measures, with behavioral values mostly between $-.02$ and $-.36$. Surprisal RSA trends are more heterogeneous: correlations with size are positive for the three N400 datasets ($r = .729$–$.795$; .558–.629 within

groups), negative for GECO and OneStop go-past times ($r = -.669$ and $-.813$), and positive for UCL self-paced reading ($r = .507$). These are descriptive associations, not estimates of a universal scaling law (Appendix H.5).

Finally, the differences between the direct-prediction and representational fronts raise the question of how closely the two kinds of evaluation agree at the level of individual datasets. We therefore correlated, across models, each dataset's direct-prediction score with its RSA score. Across-model correlations between embedding prediction scores and embedding RSA are positive in all eight available neural datasets ($r = .275$–$.929$), but vary in sign for behavior. They are positive for Provo ($r = .649$–$.678$ across its measures), negative for OneStop first-pass and total dwell times ($r = -.342$ and $-.412$), and close to zero for some other measures. Surprisal comparisons vary in both domains. Among neural datasets, correlations are about .75 for Nieuwland and Federmeier 2007, but close to zero or negative for Pereira (.019), MEG-MASC (.138) and Broderick ($-.119$). Agreement between direct prediction and RSA is thus high only for embeddings on most neural datasets, and even there it is modest for MEG-MASC (.275); for behavior, and for surprisal in both domains, it varies in sign. Consequently, good direct prediction does not guarantee the same ranking under representational evaluation. Differences in response construction and coverage also contribute to this comparison.

5 Discussion and Limitations

5.1 Discussion

Our results show that human-inspired constraints are associated with behavioral and neural alignment in different ways, rather than improving both. For contextual embeddings, a developmental data budget is associated with better behavioral prediction but not neural prediction, while memory constraints are associated with worse neural prediction and no reliable behavioral gain; the joint model shows that both differences between domains are reliable. Combining both constraints cancels the behavioral advantage of the data budget, and within memory-limited architectures, training on less data does not improve behavioral prediction. The data advantage is therefore a contrast between BabyLM-trained standard architectures and web-scale LLMs rather than an effect of the amount of data alone, and the neural memory penalty is concentrated in forget-gate and distance-biased attention rather than shared by all memory mechanisms. For surprisal, the behavioral data advantage is weaker and the neural constraint coefficients depend on datasets with unstable noise ceilings, so we do not interpret them. These findings suggest that predicting human reading behavior and neural responses places different demands on language models.

Model scale further highlights this dissociation. Larger models achieve higher neural prediction scores from embeddings, consistent with scaling accounts of representational convergence (Huh et al., 2024; Shain et al., 2024). This association is weaker once dependence between models is modeled, and it should be interpreted with caution: AlKhamissi et al. (2025) find that model size is not a reliable predictor of brain alignment once the number of features is controlled, and our ridge encoders have more features for larger models. They also find that alignment with the language network saturates early in training, and that brain alignment stops tracking next-word prediction and behavioral alignment once models surpass human proficiency, with a negative correlation between brain and behavioral alignment late in training for larger models. This divergence parallels the opposite size associations we observe for neural embedding prediction and behavioral surprisal prediction, although our models differ in size and architecture rather than in training stage. However, larger size is negatively associated with behavioral surprisal prediction after controlling for perplexity, consistent with previous reports that scaling does not necessarily improve behavioral alignment (Oh & Schuler, 2023b;a; Kuribayashi et al., 2021; de Varda et al., 2026). One possibility is that smaller or constrained models retain information particularly relevant to human processing difficulty, whereas larger models better predict neural responses. However, these associations do not establish an underlying mechanism.

Differences across evaluation methods also qualify model rankings. Direct encoding, RSA, and RDM regression capture distinct aspects of correspondence and do not always favor the same models. In particular, memory-constrained models occupy the neural end of the surprisal RDM-regression frontier despite their lower direct neural prediction scores, although neural scores on that frontier are close to or below the lexical baseline, so this reflects a ranking rather than good absolute

prediction. Together, these results suggest that neither model scale nor a single evaluation metric provides a complete characterization of human-language alignment.

5.2 Limitations

Our comparisons are observational and cannot establish causal effects of memory or data constraints. Model families differ in architecture, training objectives, and corpora, and the constraint groups have limited overlap in size and perplexity. Consequently, regression adjustment may involve extrapolation. We model the dependence between models of one lineage, between scores of one model, and between units of one corpus with random effects, and we check the conclusions with a lineage cluster bootstrap and several exclusion and outlier analyses. These checks matter: a specification with only a dataset random intercept yields much smaller standard errors, and several associations it suggests, including a behavioral advantage of memory constraints and all neural surprisal constraint effects, do not survive them. Each constrained group still contains only five or six lineages, so even the robust associations rest on a small number of independent architectures and training pipelines.

Our evaluation measures also have different assumptions and coverage. Embedding encoders do not explicitly control for lexical variables, unlike surprisal and RDM-regression analyses. Noise-ceiling estimates have finite reliability, and normalized neural scores can vary substantially across datasets. Moreover, representational analyses cover fewer neural datasets, and differences between evaluation methods cannot be attributed to a single factor.

Finally, developmental data restrictions and the tested memory mechanisms capture only limited aspects of human language learning and cognition. Our conclusions are restricted to the evaluated model families, final-layer representations, and predominantly English datasets with limited participant or recording coverage. Most models remain below the estimated noise ceilings, leaving substantial room for improvement. Predictive correspondence alone does not establish shared representations or computational mechanisms.

Overall, a developmental data budget is associated with better prediction of human reading behavior without better prediction of neural responses, and memory constraints with worse neural prediction, emphasizing the importance of evaluating both targets with complementary measures.

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A Noise Ceilings and Normalization

Raw correlations and log-likelihood improvements are not comparable across datasets: a Pearson r of 0.3 might be close to the achievable maximum in a noisy single-trial intracranial recording and far from it in a clean, heavily averaged fMRI dataset. The direct-prediction analyses therefore divide every dataset-unit score by a reference derived from the reliability of the human data, both in the Pareto composites and in Equation 2. They do not standardize outcomes by the spread of scores across models. Predictive composites take the median across dataset units within each response domain; regressions retain individual model–unit rows. Thus frontier membership and regression coefficients summarize different aspects of the data. We estimate two kinds of reference, one for each predictor track.

A.1 Correlation ceilings: between-subject split-half reliability

For every dataset where a model's score is a correlation between predicted and observed responses (at the level of an electrode, voxel, ROI, or channel), we estimate a noise ceiling from the split-half reliability of the human data, following van Bree et al. (2025).

Our datasets typically present each item once to each participant, so there is no within-subject repetition to split on. We instead split *across* subjects: for a given dataset and response unit, we repeatedly draw two random, equally sized halves of the subject pool, average each half's response per item, and correlate the two half-averages across items to obtain a split-half correlation r_{half}, the group-average variant of split-half reliability that van Bree et al. describe for designs without stimulus repetitions. We apply the Spearman–Brown correction to extrapolate from a half-sample to the full sample,

$$r_{\text{SB}} = \frac{2\, r_{\text{half}}}{1 + r_{\text{half}}},$$

and average r_{SB} over many random subject splits to stabilize the estimate.

It is tempting to treat r_{SB} itself as the ceiling for a model's correlation with the data, but this is the mistake van Bree et al. warn against: r_{SB} is a correlation between two noisy measurements, so by the classical correction-for-attenuation logic it bounds explained variance (R^2), not the correlation coefficient itself. A model that perfectly recovers the true signal is not itself noisy, so its correlation with the (still noisy) data is attenuated only once rather than twice, giving

$$r_{\max} = \sqrt{r_{\text{SB}}}.$$

We therefore divide every model correlation by $\sqrt{r_{\text{SB}}}$. Using r_{SB} as the ceiling for a raw correlation would systematically understate how far a model is from ceiling, especially at the low-to-intermediate reliability that characterizes most of our neural datasets. Ceilings are computed separately per response unit, using the same correlation family (Pearson or Spearman) as the corresponding model score, and response averaging and regional aggregation follow each dataset's scoring procedure. For Le Petit Prince the reference is a precomputed intersubject-correlation conversion. A correlation ratio is not explained variance, and differences between ratios should not be interpreted as percentage changes in predictive accuracy.

A.2 Delta log-likelihood ceilings: the split-half design

For the surprisal track, where the quantity of interest is the log-likelihood improvement $\Delta\ell$ that surprisal contributes over a nuisance-only baseline, a correlation-based ceiling does not directly apply. We instead adapt the split-half procedure that Timkey et al. (2026) use to contextualize their filler-sentence regression models: we ask how much $\Delta\ell$ a predictor could achieve given only the information available to a single word-level scalar like surprisal, i.e. one value per item, with no participant- or trial-level access.

We construct this predictor directly from the data. For a given dataset, we randomly split the subject pool into two equal halves, A and B. Within A, we compute the per-item mean response (averaged across subjects and, where relevant, electrodes), substitute it for surprisal in a mixed-effects model otherwise identical to the real full model, and fit it to half B. Comparing this model's held-out log likelihood with that of the baseline model (identical formula, predictor omitted) on the same B data

gives one estimate of the achievable $\Delta\ell$. We swap the halves, using B's per-item means to predict A, sum the $\Delta\ell$ from both directions, and repeat over random splits (typically 15), reporting the mean per-observation $\Delta\ell$ as the reference. Two details carry over from the main fitting procedure: references are computed out of sample, using the same item or chunk train–test split as the corresponding real analysis, and the reference model's random-effects structure mirrors the real model's for that dataset, so the only difference between the two is the substitution of an empirical per-item mean for the model-derived surprisal. For datasets fit per region (Podcast, Pereira), the reference is computed per region and each region's score is normalized by its own reference before the regions are combined.

Because the substitute predictor is a finite-sample group average rather than a noise-free oracle, it is not an upper bound in the strict sense. The reference can be small, uncertain, or exceeded by a model scored against a fuller response average, particularly when there are few subjects per item or the achievable $\Delta\ell$ is very small. Ratios above one are therefore retained and do not establish superhuman prediction. The same issue applies to the split-half references used in RDM regression (Appendix H). Neither these ratios nor their cross-dataset medians are fractions of explainable variance.

A.3 Reported ceilings and score ranges

Tables 4 and 5 report every reference used in the final-layer direct-prediction scores, with the largest normalized score among the 118 models and the number of models above the reference. For the embedding track, no model exceeds its ceiling; the best models reach between about 12% and 78% of the ceiling, and are furthest from it for Natural Stories fMRI and the continuous neural recordings (MEG-MASC, Podcast, Broderick). For the surprisal track, four references (UCL self-paced reading, Broderick, Federmeier et al. 2007, Hubbard et al. 2019; marked †) are exceeded by many models or have a split-to-split SD larger than half their value; they rest on few subjects per item or on a very small achievable $\Delta\ell$, and normalized scores on them should be interpreted with that in mind. A low reference indicates that the human measurement itself carries limited reliable signal at the item level.

Table 6 reports the resulting normalized scores. Neural surprisal scores have markedly different ranges across datasets: Federmeier 2007 spans approximately -3.44 to 6.98, whereas MEG-MASC spans approximately $-.013$ to $.007$. These differences motivate caution about pooled domain coefficients: a dataset random intercept absorbs location differences, not differences in residual scale or uncertain denominators. Appendix E therefore refits the group analysis without the flagged units, with winsorized scores, on a within-unit robust z-score, and leaving out one unit at a time.

Table 4: Correlation references used to normalize the final-layer embedding scores. Maximum is the largest normalized score among models with a finite result, not a selected layer. Pearson and Rank indicate Pearson- and Spearman-based split-half references; ISC denotes the precomputed intersubject-correlation conversion. Above counts normalized scores greater than one. Rank references are reliability normalizers, not exact Pearson ceilings. See Appendix A.1 for aggregation and target-matching qualifications.

Dataset / measure	Reference	Type	N	Maximum	Above
GECO FP	0.627	Pearson	118	0.587	0/118
GECO GP	0.529	Pearson	118	0.543	0/118
MECO FP	0.899	Pearson	118	0.650	0/118
MECO GP	0.684	Pearson	118	0.453	0/118
MECO SP	0.930	Pearson	118	0.737	0/118
Natural Stories SPR	0.914	Pearson	118	0.450	0/118
OneStop FP	0.829	Pearson	118	0.422	0/118
OneStop GP	0.776	Pearson	118	0.310	0/118
OneStop SP	0.888	Pearson	118	0.523	0/118
Provo FP	0.933	Pearson	118	0.604	0/118
Provo GP	0.919	Pearson	118	0.403	0/118
Provo SP	0.943	Pearson	118	0.608	0/118
UCL SPR	0.689	Pearson	118	0.470	0/118

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Table 4 – continued

Dataset / measure	Reference	Type	N	Maximum	Above
ZuCo NR	0.819	Pearson	118	0.741	0/118
ZuCo SR	0.819	Pearson	118	0.757	0/118
ZuCo TSR	0.819	Pearson	118	0.598	0/118
Le Petit Prince fMRI	0.912	ISC	118	0.447	0/118
Natural Stories fMRI	0.628	Pearson	118	0.120	0/118
Pereira fMRI	0.507	Pearson	118	0.238	0/118
Tuckute fMRI	0.480	Pearson	118	0.299	0/118
Fedorenko intracranial	0.711	Rank	118	0.291	0/118
Broderick EEG	0.108	Pearson	118	0.148	0/118
MEG-MASC	0.097	Pearson	118	0.129	0/118
Nieuwland N400	0.887	Rank	118	0.425	0/118
Federmeier 2007 N400	0.693	Rank	118	0.746	0/118
Hubbard 2019 N400	0.708	Rank	118	0.777	0/118
Podcast ECoG	0.398	Pearson	118	0.157	0/118

Table 5: Split-half likelihood references used in final-layer surprisal normalization, in $\Delta\ell$ per observation. SD is variability across splits or iterations, not a standard error; – means the aggregate file supplies no such value. Splits reports successful/attempted draws where available. Above is the number of finite model scores exceeding the reference, and Maximum is the largest normalized score. A dagger flags split SD/reference $> .5$ or more than 10% of models above one; this is a descriptive diagnostic, not an exclusion rule. Pereira split counts apply to each of its two ROIs; region-specific references are detailed in the text.

Dataset / measure	Reference	SD	Splits	Above	Maximum
GECO FP	8.66×10^{-3}	3.59×10^{-4}	10/10	0/118	0.272
GECO GP	4.16×10^{-3}	4.30×10^{-4}	15/15	0/118	0.183
MECO FP	2.34×10^{-2}	3.68×10^{-3}	15/15	0/118	0.172
MECO GP	7.97×10^{-3}	1.86×10^{-3}	15/15	0/118	0.089
MECO SP	3.79×10^{-2}	2.56×10^{-3}	15/15	0/118	0.190
Natural Stories SPR	2.38×10^{-2}	9.14×10^{-4}	15/15	0/118	0.395
OneStop FP	1.29×10^{-2}	5.59×10^{-4}	15/15	0/118	0.379
OneStop GP	1.43×10^{-2}	8.90×10^{-4}	15/15	0/118	0.091
OneStop SP	2.36×10^{-2}	1.39×10^{-3}	15/15	0/118	0.373
Provo FP	3.35×10^{-2}	1.50×10^{-3}	15/15	0/118	0.138
Provo GP	3.60×10^{-2}	1.60×10^{-3}	15/15	0/118	0.064
Provo SP	4.57×10^{-2}	1.02×10^{-3}	15/15	0/118	0.194
UCL SPR†	2.36×10^{-3}	2.61×10^{-4}	15/15	51/118	1.503
ZuCo NR	2.78×10^{-2}	2.25×10^{-3}	10/10	0/118	0.311
ZuCo SR	3.14×10^{-2}	1.55×10^{-3}	15/15	0/118	0.454
ZuCo TSR	1.25×10^{-2}	2.39×10^{-3}	15/15	0/118	0.629
Le Petit Prince fMRI	1.01×10^{-2}	8.79×10^{-4}	15/15	0/118	0.057
Natural Stories fMRI	2.97×10^{-2}	3.94×10^{-3}	15/15	0/118	0.134
Pereira fMRI	2.37×10^{-2}	–	29/30	0/118	0.640
Tuckute fMRI	9.52×10^{-3}	1.32×10^{-3}	10/10	0/118	0.248
Fedorenko intracranial	8.32×10^{-3}	1.62×10^{-3}	10/10	4/118	1.150
Podcast ECoG	6.19×10^{-3}	–	–	0/118	0.118
Broderick EEG†	1.00×10^{-4}	7.00×10^{-5}	60/60	42/118	2.293
MEG-MASC	7.31×10^{-4}	1.78×10^{-4}	15/15	0/118	0.007
Nieuwland N400	1.25×10^{-2}	1.20×10^{-3}	30/30	0/118	0.342
Federmeier 2007 N400†	2.61×10^{-3}	2.75×10^{-3}	15/15	67/118	6.976
Hubbard 2019 N400†	2.22×10^{-3}	1.21×10^{-3}	15/15	49/118	2.607

Table 6: Normalized final-layer scores of the direct-prediction analyses. Each entry is minimum / median / maximum across available models in that unit. These ranges describe the input tables before any further complete-case filtering for the regression.

Dataset unit	Embeddings	Surprisal
Broderick EEG	0.005 / 0.111 / 0.148	-0.077 / 0.341 / 2.293
Fedorenko intracranial	0.078 / 0.247 / 0.291	-0.239 / 0.155 / 1.150
GECO FP	0.408 / 0.543 / 0.587	0.021 / 0.205 / 0.272
GECO GP	0.370 / 0.483 / 0.543	0.002 / 0.108 / 0.183
Le Petit Prince fMRI	0.038 / 0.383 / 0.447	-0.026 / 0.010 / 0.057
MECO FP	0.471 / 0.587 / 0.650	-0.004 / 0.069 / 0.172
MECO GP	0.266 / 0.377 / 0.453	-0.182 / 0.003 / 0.089
MECO SP	0.535 / 0.648 / 0.737	0.005 / 0.070 / 0.190
MEG-MASC	0.026 / 0.097 / 0.129	-0.013 / -0.004 / 0.007
Natural Stories SPR	0.193 / 0.344 / 0.450	0.151 / 0.263 / 0.395
Nieuwland N400	-0.362 / 0.214 / 0.425	-0.608 / 0.028 / 0.342
Natural Stories fMRI	0.015 / 0.068 / 0.120	0.024 / 0.064 / 0.134
OneStop FP	0.282 / 0.383 / 0.422	0.007 / 0.214 / 0.379
OneStop GP	0.212 / 0.273 / 0.310	0.011 / 0.057 / 0.091
OneStop SP	0.339 / 0.464 / 0.523	0.017 / 0.231 / 0.373
Pereira fMRI	0.116 / 0.203 / 0.238	-0.157 / 0.219 / 0.640
Podcast ECoG	0.030 / 0.125 / 0.157	0.013 / 0.088 / 0.118
Provo FP	0.394 / 0.517 / 0.604	0.012 / 0.068 / 0.138
Provo GP	0.234 / 0.341 / 0.403	0.001 / 0.033 / 0.064
Provo SP	0.407 / 0.527 / 0.608	0.007 / 0.131 / 0.194
Federmeier 2007 N400	0.149 / 0.519 / 0.746	-3.436 / 3.138 / 6.976
Hubbard 2019 N400	0.044 / 0.515 / 0.777	-1.099 / 0.743 / 2.607
Tuckute fMRI	0.025 / 0.223 / 0.299	-0.031 / 0.121 / 0.248
UCL SPR	0.205 / 0.316 / 0.470	0.075 / 0.984 / 1.503
ZuCo NR	0.615 / 0.693 / 0.741	-0.017 / 0.196 / 0.311
ZuCo SR	0.565 / 0.696 / 0.757	0.038 / 0.333 / 0.454
ZuCo TSR	0.479 / 0.555 / 0.598	0.157 / 0.319 / 0.629

B Full List of Evaluated Models

Table 7 lists every model evaluated in this benchmark, with its family, depth, parameter count, and HuggingFace repository, organized into the four groups of Section 3.2. The main comparison uses released model instances rather than intermediate training checkpoints. The four groups contain 59, 24, 11, and 24 models, respectively, for Standard LLMs, Memory-Constrained, BabyLM, and Constrained-BabyLM. The direct-prediction regressions retain 115 models after requiring usable scores and perplexity. Architectural depth metadata can include an embedding state in some implementations; the analysis consistently uses the final representation available in the corresponding evaluation.

The Memory-Constrained and Constrained-BabyLM groups (identical small architectures, web-scale vs. BabyLM-scale training data) cover several mechanisms: architectures that bias attention toward nearby tokens through a fixed distance penalty, ALiBi (Press et al., 2021; de Varda & Marelli, 2024; Clark et al., 2025); architectures that replace softmax attention with a content-sensitive forget gate (Lin et al., 2025) or a stick-breaking process (Tan et al., 2025); and hard local n-gram windows that can only attend to the two, three, or five preceding tokens regardless of the input (Bu et al., 2026). The web-scale group additionally contains CBR-RNN and its variant CBR-RNN-M, a small recurrent architecture with a bounded context representation (Timkey & Linzen, 2023), and larger web-scale models built on the same mechanisms (BLOOM with ALiBi, the Forgetting Transformer, and MoLM). The small architectures and their BabyLM-scale counterparts were obtained from the public HuggingFace checkpoints released alongside Bu et al. (2026). The BabyLM group spans submissions and baselines from the BabyLM shared tasks, including the organizers' baselines (Charpentier et al., 2025), a masked-language-modeling variant (Edman & Fraser, 2025), an instruction-tuned small model (Capone et al., 2025), a mixture-of-experts architecture (Tapaninaho, 2025), and a model trained with interactive storytelling feedback (Mayer Martins et al., 2025), alongside the earlier BabyLM challenges (Warstadt et al., 2023a;b; Hu et al., 2024), all obtained from their public HuggingFace repositories. The Standard LLM group spans off-the-shelf families with no human-inspired constraint on either architecture or training data: GPT-2 (Radford et al., 2019), GPT-Neo (Black et al., 2021), OPT (Zhang et al., 2022), Mamba and Mamba-2 (Gu & Dao, 2024), RWKV-4, Pythia (Biderman et al., 2023), OLMo-2 (Team OLMo et al., 2025), Qwen3 (Yang et al., 2025), Llama-3, Ministral, and the attention baselines of the Forgetting Transformer (Lin et al., 2025), from 14 million to 66 billion parameters and including both Transformer and state-space architectures.

Table 7: All 118 models evaluated in this benchmark, grouped by family within the model groups of Section 3.2 (rules separate each fine-grained family). Memory-Constrained models trained on BabyLM-scale data (marked by `babylm` in the repository name) form the Constrained-BabyLM group; all other models listed under Memory-Constrained are trained on web-scale text. The two "FoX-release baseline" rows are the matched baselines released with FoX, with ordinary attention and no forget gate (the Pro variant keeps the output gate and key/value shift of FoX-Pro); they carry no memory constraint and are therefore counted as Standard LLMs. Parameter and layer counts follow the originating release (or the loaded config for the memory-constrained and BabyLM checkpoints; see `neuroai/models/registry.py`). Training-token/word counts are reported only where documented with confidence by the originating source; a blank cell means this was not reliably available rather than zero.

Family	HuggingFace model ID	#Params	#Layers	Tr. tokens	Ref.
Standard LLMs					
GPT-2	openai-community/gpt2	124M	13		(Radford et al., 2019)
GPT-2	openai-community/gpt2-medium	355M	25		(Radford et al., 2019)
GPT-2	openai-community/gpt2-large	774M	37		(Radford et al., 2019)

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Table 7 – continued

Family	HuggingFace model ID	#Params	#Layers	Tr. tokens	Ref.
GPT-2	openai-community/gpt2-xl	1.6B	49		(Radford et al., 2019)
GPT-Neo/J/NeoX	EleutherAI/gpt-neo-125m	125M	13		(Black et al., 2021)
GPT-Neo/J/NeoX	EleutherAI/gpt-neo-1.3B	1.3B	25		(Black et al., 2021)
GPT-Neo/J/NeoX	EleutherAI/gpt-neo-2.7B	2.7B	33		(Black et al., 2021)
GPT-Neo/J/NeoX	EleutherAI/gpt-j-6b	6B	29	402B	(Wang & Komatsuzaki, 2021)
GPT-Neo/J/NeoX	EleutherAI/gpt-neox-20b	20B	45	472B	(Black et al., 2022)
OPT	facebook/opt-125m	125M	13	180B	(Zhang et al., 2022)
OPT	facebook/opt-350m	350M	25	180B	(Zhang et al., 2022)
OPT	facebook/opt-1.3b	1.3B	25	180B	(Zhang et al., 2022)
OPT	facebook/opt-2.7b	2.7B	33	180B	(Zhang et al., 2022)
OPT	facebook/opt-6.7b	6.7B	33	180B	(Zhang et al., 2022)
OPT	facebook/opt-13b	13B	41	180B	(Zhang et al., 2022)
OPT	facebook/opt-30b	30B	49	180B	(Zhang et al., 2022)
OPT	facebook/opt-66b	66B	65	180B	(Zhang et al., 2022)
Mamba	state-spaces/mamba-130m-hf	130M	25	300B	(Gu & Dao, 2024)
Mamba	state-spaces/mamba-370m-hf	370M	49	300B	(Gu & Dao, 2024)
Mamba	state-spaces/mamba-790m-hf	790M	49	300B	(Gu & Dao, 2024)
Mamba	state-spaces/mamba-1.4b-hf	1.4B	49	300B	(Gu & Dao, 2024)
Mamba	state-spaces/mamba-2.8b-hf	2.8B	61	300B	(Gu & Dao, 2024)
Mamba-2	AntonV/mamba2-130m-hf	130M	25	300B	(Dao & Gu, 2024)
Mamba-2	AntonV/mamba2-370m-hf	370M	49	300B	(Dao & Gu, 2024)
Mamba-2	AntonV/mamba2-780m-hf	780M	49	300B	(Dao & Gu, 2024)
Mamba-2	AntonV/mamba2-1.3b-hf	1.3B	49	300B	(Dao & Gu, 2024)
Mamba-2	AntonV/mamba2-2.7b-hf	2.7B	61	300B	(Dao & Gu, 2024)
Pythia	EleutherAI/pythia-14m-deduped	14M	6	300B	(Biderman et al., 2023)
Pythia	EleutherAI/pythia-31m-deduped	31M	6	300B	(Biderman et al., 2023)
Pythia	EleutherAI/pythia-70m-deduped	70M	6	300B	(Biderman et al., 2023)
Pythia	EleutherAI/pythia-160m-deduped	160M	12	300B	(Biderman et al., 2023)
Pythia	EleutherAI/pythia-410m-deduped	410M	24	300B	(Biderman et al., 2023)
Pythia	EleutherAI/pythia-1b-deduped	1B	17	300B	(Biderman et al., 2023)
Pythia	EleutherAI/pythia-1.4b-deduped	1.4B	25	300B	(Biderman et al., 2023)
Pythia	EleutherAI/pythia-2.8b-deduped	2.8B	33	300B	(Biderman et al., 2023)
Pythia	EleutherAI/pythia-6.9b-deduped	6.9B	33	300B	(Biderman et al., 2023)
Pythia	EleutherAI/pythia-12b-deduped	12B	37	300B	(Biderman et al., 2023)
OLMo-2	allenai/OLMo-2-0425-1B	1B	17	4.0T	(Team OLMo et al., 2025)
OLMo-2	allenai/OLMo-2-1124-7B	7B	32	3.9T	(Team OLMo et al., 2025)
OLMo-2	allenai/OLMo-2-1124-13B	13B	40	5.0T	(Team OLMo et al., 2025)
Qwen3	Qwen/Qwen3-0.6B-Base	600M	29	36T	(Yang et al., 2025)
Qwen3	Qwen/Qwen3-1.7B-Base	1.7B	29	36T	(Yang et al., 2025)
Qwen3	Qwen/Qwen3-4B-Base	4B	37	36T	(Yang et al., 2025)
Qwen3	Qwen/Qwen3-8B-Base	8B	37	36T	(Yang et al., 2025)
Qwen3	Qwen/Qwen3-14B-Base	14B	41	36T	(Yang et al., 2025)
Llama-3	meta-llama/Llama-3.2-1B	1B	17	9T	
Llama-3	meta-llama/Llama-3.2-3B	3B	29	9T	

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Table 7 – continued

Family	HuggingFace model ID	#Params	#Layers	Tr. tokens	Ref.
Llama-3	meta-llama/Llama-3.1-8B	8B	33	15T	
Ministral	mistralai/Ministral-3-3B-Base-2512	3B	29		
Ministral	mistralai/Ministral-3-8B-Base-2512	8B	37		
Ministral	mistralai/Ministral-3-14B-Base-2512	14B	41		
RWKV-4	RWKV/rwkv-4-169m-pile	169M	13		(Peng et al., 2023)
RWKV-4	RWKV/rwkv-4-430m-pile	430M	25		(Peng et al., 2023)
RWKV-4	RWKV/rwkv-4-1b5-pile	1.5B	25		(Peng et al., 2023)
RWKV-4	RWKV/rwkv-4-3b-pile	3B	33		(Peng et al., 2023)
RWKV-4	RWKV/rwkv-4-7b-pile	7B	33		(Peng et al., 2023)
RWKV-4	RWKV/rwkv-4-14b-pile	14B	41		(Peng et al., 2023)
FoX-release baseline	zhixuan-lin/transformer-llama-760m-longcrawl64-48b	760M	25	48B	(Lin et al., 2025)
FoX-release baseline	zhixuan-lin/transformer-pro-760m-longcrawl64-48b	760M	25	48B	(Lin et al., 2025)
Memory-Constrained					
CBR-RNN	cclark/cbr-rnn	17.4M	1		(Timkey & Linzen, 2023)
CBR-RNN	cclark/cbr-rnn-m	17.4M	1		(Timkey & Linzen, 2023)
Stick-Breaking	Lanni-ni/stickbreaking_pile_2layer	27.8M	2		(Tan et al., 2025)
Stick-Breaking	Lanni-ni/stickbreaking_pile_4layer	48.1M	4		(Tan et al., 2025)
Stick-Breaking	Lanni-ni/stickbreaking_babylm_10m_2layer	27.8M	2	∼10M w.	(Tan et al., 2025)
Stick-Breaking	Lanni-ni/stickbreaking_babylm_10m_4layer	48.1M	4	∼10M w.	(Tan et al., 2025)
Stick-Breaking	Lanni-ni/stickbreaking_babylm_100m_2layer	27.8M	2	∼100M w.	(Tan et al., 2025)
Stick-Breaking	Lanni-ni/stickbreaking_babylm_100m_4layer	48.1M	4	∼100M w.	(Tan et al., 2025)
ALiBi	Lanni-ni/alibi_pile_2layer	27.4M	2		(Press et al., 2021)
ALiBi	Lanni-ni/alibi_pile_4layer	45.7M	4		(Press et al., 2021)
ALiBi	Lanni-ni/alibi_babylm_10m_2layer	27.4M	2	∼10M w.	(Press et al., 2021)
ALiBi	Lanni-ni/alibi_babylm_10m_4layer	45.7M	4	∼10M w.	(Press et al., 2021)
ALiBi	Lanni-ni/alibi_babylm_100m_2layer	27.4M	2	∼100M w.	(Press et al., 2021)
ALiBi	Lanni-ni/alibi_babylm_100m_4layer	45.7M	4	∼100M w.	(Press et al., 2021)
Forgetting-Transf.	Lanni-ni/forgetting_pile_2layer	27.4M	2		(Lin et al., 2025)
Forgetting-Transf.	Lanni-ni/forgetting_pile_4layer	45.7M	4		(Lin et al., 2025)
Forgetting-Transf.	Lanni-ni/forgetting_babylm_10m_2layer	27.4M	2	∼10M w.	(Lin et al., 2025)
Forgetting-Transf.	Lanni-ni/forgetting_babylm_10m_4layer	45.7M	4	∼10M w.	(Lin et al., 2025)
Forgetting-Transf.	Lanni-ni/forgetting_babylm_100m_2layer	27.4M	2	∼100M w.	(Lin et al., 2025)

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Table 7 – continued

Family	HuggingFace model ID	#Params	#Layers	Tr. tokens	Ref.
Forgetting-Transf.	Lanni-ni/forgetting_babylm_100m_41ayer	45.7M	4	∼100M w.	(Lin et al., 2025)
Hard 2-gram	Lanni-ni/hard_2gram_pile_2layer	15M	2		(Bu et al., 2026)
Hard 2-gram	Lanni-ni/hard_2gram_pile_4layer	28.8M	4		(Bu et al., 2026)
Hard 2-gram	Lanni-ni/hard_2gram_babylm_10m_2layer	15M	2	∼10M w.	(Bu et al., 2026)
Hard 2-gram	Lanni-ni/hard_2gram_babylm_10m_4layer	28.8M	4	∼10M w.	(Bu et al., 2026)
Hard 2-gram	Lanni-ni/hard_2gram_babylm_100m_21ayer	15M	2	∼100M w.	(Bu et al., 2026)
Hard 2-gram	Lanni-ni/hard_2gram_babylm_100m_41ayer	28.8M	4	∼100M w.	(Bu et al., 2026)
Hard 3-gram	Lanni-ni/hard_3gram_pile_2layer	15M	2		(Bu et al., 2026)
Hard 3-gram	Lanni-ni/hard_3gram_pile_4layer	28.8M	4		(Bu et al., 2026)
Hard 3-gram	Lanni-ni/hard_3gram_babylm_10m_2layer	15M	2	∼10M w.	(Bu et al., 2026)
Hard 3-gram	Lanni-ni/hard_3gram_babylm_10m_4layer	28.8M	4	∼10M w.	(Bu et al., 2026)
Hard 3-gram	Lanni-ni/hard_3gram_babylm_100m_21ayer	15M	2	∼100M w.	(Bu et al., 2026)
Hard 3-gram	Lanni-ni/hard_3gram_babylm_100m_41ayer	28.8M	4	∼100M w.	(Bu et al., 2026)
Hard 5-gram	Lanni-ni/hard_5gram_pile_2layer	15M	2		(Bu et al., 2026)
Hard 5-gram	Lanni-ni/hard_5gram_pile_4layer	28.8M	4		(Bu et al., 2026)
Hard 5-gram	Lanni-ni/hard_5gram_babylm_10m_2layer	15M	2	∼10M w.	(Bu et al., 2026)
Hard 5-gram	Lanni-ni/hard_5gram_babylm_10m_4layer	28.8M	4	∼10M w.	(Bu et al., 2026)
Hard 5-gram	Lanni-ni/hard_5gram_babylm_100m_21ayer	15M	2	∼100M w.	(Bu et al., 2026)
Hard 5-gram	Lanni-ni/hard_5gram_babylm_100m_41ayer	28.8M	4	∼100M w.	(Bu et al., 2026)
BLOOM	bigscience/bloom-560m	560M	25		(BigScience Workshop, 2022)
BLOOM	bigscience/bloom-1b1	1.1B	25		(BigScience Workshop, 2022)
BLOOM	bigscience/bloom-1b7	1.7B	25		(BigScience Workshop, 2022)
BLOOM	bigscience/bloom-3b	3B	31		(BigScience Workshop, 2022)
BLOOM	bigscience/bloom-7b1	7.1B	31		(BigScience Workshop, 2022)
FoX (Forgetting Transf.)	zhixuan-lin/fox-llama-760m-longcrawl64-48b	760M	25	48B	(Lin et al., 2025)
FoX (Forgetting Transf.)	zhixuan-lin/fox-pro-760m-longcrawl64-48b	760M	25	48B	(Lin et al., 2025)
MoLM	ibm/MoLM-350M-4B	350M	25		(Shen et al., 2023)
MoLM	ibm/MoLM-700M-4B	700M	25		(Shen et al., 2023)
MoLM	ibm/MoLM-700M-8B	700M	49		(Shen et al., 2023)
BabyLMs					
BabyLM	colinglab/CLASS_IT-140M	140M	13		(Capone et al., 2025)

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Table 7 – continued

Family	HuggingFace model ID	#Params	#Layers	Tr. tokens	Ref.
BabyLM	llm-slice/blm-gpt2s-90M-s42_901M-s42_submission	90M	13		
BabyLM	Jtapsa/moep	80M	13		(Tapaninaho, 2025)
BabyLM	BabyLM-community/BabyLM-2026-Baseline-GPT2-Strict	98.4M	13	∼100M w.	(Charpentier et al., 2025)
BabyLM	BabyLM-community/BabyLM-2026-Baseline-GPT2-Strict-Small	98.4M	13	∼10M w.	(Charpentier et al., 2025)
BabyLM	BabyLM-community/babylm-baseline-100m-gpt-bert-causal-focus	119.5M	13	∼100M w.	(Edman & Fraser, 2025)
BabyLM	BabyLM-community/babylm-baseline-100m-gpt-bert-mixed	119.5M	13	∼100M w.	(Edman & Fraser, 2025)
BabyLM	BabyLM-community/babylm-baseline-100m-gpt-bert-masked-focus	119.5M	13	∼100M w.	(Edman & Fraser, 2025)
BabyLM	BabyLM-community/babylm-baseline-10m-gpt-bert-causal-focus	33M	13	∼10M w.	(Edman & Fraser, 2025)
BabyLM	BabyLM-community/babylm-baseline-10m-gpt-bert-mixed	33M	13	∼10M w.	(Edman & Fraser, 2025)
BabyLM	BabyLM-community/babylm-baseline-10m-gpt-bert-masked-focus	33M	13	∼10M w.	(Edman & Fraser, 2025)

C Full List of Evaluated Datasets

Table 8 lists every dataset, grouped by modality, corresponding to the behavioral and neural categories of Section 3.3. The 18 datasets yield 27 prediction units. Behavioral measures from one corpus are separate outcomes, not independent replications. The representational fronts use 16 behavioral measures and eight neural datasets. They exclude Le Petit Prince and Natural Stories fMRI, which have no corresponding RSA scores, and Fedorenko, which lacks a usable common RDM ceiling for these composites.

Table 8: All datasets evaluated in this benchmark, grouped by modality. N items reports the unit each dataset's items are counted in (sentences, passages, screens, stories, electrodes, ...), since this varies by design. Eye-tracking measures: FP = first-pass/gaze duration, GP = go-past time, SP = total dwell time. N subj. reflects each study's usable sample after its own quality-control exclusions.

Dataset	Modality	Subj.	Items	Measure(s)	Ref.
Self-Paced Reading					
Natural Stories SPR	SPR	180	10 stories (10,256 word pos.)	raw RT	(Futrell et al., 2021)
UCL Reading	SPR	117	361 sentences	raw RT	(Frank et al., 2013)
Eye-Tracking					
GECO	Eye-tracking	14	588 screens	FP, GP	(Cop et al., 2017)
MECO (English)	Eye-tracking	46	12 passages	SP, FP, GP	(Siegelman et al., 2022)
Provo	Eye-tracking	84	55 texts	SP, FP, GP	(Luke & Christianson, 2018)
OneStop	Eye-tracking	360	330 passages	SP, FP, GP	(Berzak et al., 2025)

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Table 8 – continued

Dataset	Modality	Subj.	Items	Measure(s)	Ref.
ZuCo Reading	Eye-tracking	12	SR/NR/TSR: 400/300/407 sent.	TRT	(Hollenstein et al., 2018)
N400 / ERP					
Nieuwland	EEG (ERP)	334	9-lab RSVP replication	N400	(Nieuwland et al., 2018)
Federmeier et al. (2007)	EEG (ERP)	32	564 items	N400	(Federmeier et al., 2007; Szewczyk & Federmeier, 2022)
Hubbard et al. (2019)	EEG (ERP)	32	192 items	N400	(Szewczyk & Federmeier, 2022)
Continuous EEG					
Broderick	EEG	19	20 segs. ($\sim$196s each)	EEG (1–8Hz)	(Broderick et al., 2018)
MEG					
Gwilliams (MEG-MASC)	MEG	27	4 stories $\times$ 2 sess.	continuous MEG	(Gwilliams et al., 2022)
fMRI					
Le Petit Prince	fMRI	49	9 sections (audio)	BOLD	(Li et al., 2022)
Natural Stories fMRI	fMRI	78	8 narratives (26 fROIs)	BOLD	(Shain et al., 2022)
Tuckute	fMRI	10	2,000 sentences (6 ROIs)	BOLD	(Tuckute et al., 2024)
Pereira	fMRI	$\leq$10	627 sentences (12 ROIs)	BOLD	(Pereira et al., 2018)
iEEG / ECoG					
Podcast	ECoG	9	1 story, >5,000 w. (1,330 electr.)	high-gamma	(Zada et al., 2025)
Fedorenko2016	iEEG	5	52 sentences (416 words)	iEEG (scalar)	(Fedorenko et al., 2016)

D Implementation Details

Data provenance. All datasets were obtained from their public OpenNeuro or OSF repositories under the terms specified therein.

Feature extraction. All language models are loaded in bfloat16 and evaluated as frozen feature extractors, with no gradient updates. For each word, final-layer hidden states are mean-pooled over its constituent subword tokens, and a word's representation is computed with its full preceding sentence or discourse context in the forward pass rather than in isolation. Word surprisal is the sum of the subword-token surprisals of the model's own output distribution. Dataset-specific preprocessing, target aggregation, split definitions, regularization settings, and nuisance specifications are identical across the compared models.

Ridge regression. For every dataset, nested cross-validation selects the ridge penalty from a log-spaced grid via an inner search, with predictors standardized by a robust (median/IQR) scaler fit on the training fold only, and outer folds grouped so that no sentence, item, story, or run straddles the train/test boundary. The grid and fold count are tuned per dataset to its sample size and predictor dimensionality rather than fixed globally; Table 9 lists the configuration for every dataset. We report the mean out-of-fold correlation across the outer folds.

Table 9: Ridge regression configuration for the ridge encoder of each dataset (embedding track): log-spaced α grid (number of grid points in parentheses), outer cross-validation fold count, and the grouping unit used so that no fold splits correlated observations. GroupKFold groups on the listed unit unless marked LOSO (leave-one-unit-out, i.e. fold count equals unit count).

Dataset	α grid (points)	Folds	Grouping unit
GECO (FP/GP)	10^{1}–10^{10} (10)	5	screen
MECO (FP/GP/SP)	$10^{0.5}$–10^{10} (12)	5	passage
Provo (FP/GP/SP)	10^{0}–10^{10} (11)	5	text
UCL Reading	$10^{1.5}$–10^{9} (9)	5	sentence
OneStop (FP/GP/SP)	10^{0}–10^{10} (11)	5	passage
Natural Stories SPR	10^{-1}–10^{5} (7)	10 (LOSO)	story
Natural Stories fMRI	10^{-2}–10^{4} (6)	8 (LOSO)	story
ZuCo Reading (NR)	$10^{2.5}$–10^{9} (8)	5	sentence
ZuCo Reading (SR/TSR)	10^{2}–10^{6} (8)	5	sentence
Nieuwland	$10^{3.5}$–10^{10} (6)	10	sentence item
Federmeier et al. (2007), Hubbard et al. (2019)	$10^{1.5}$–$10^{5.25}$ (7)	10	sentence frame
Tuckute	10^{2}–10^{7} (5)	4	sentence
Broderick	$10^{3.5}$–10^{8} (5)	3	audiobook segment
MEG-MASC	$10^{3.5}$–10^{8} (5)	4 (LOSO)	story
Le Petit Prince	10^{3}–$10^{5.3}$ (8)	9 (LOSO)	section
Podcast	$10^{3.5}$–10^{8} (5)	3	contiguous time chunk
Fedorenko2016	$10^{3.5}$–10^{8} (5)	5	sentence
Pereira	10^{1}–10^{6} (8)	5	passage

Temporal response function (TRF) datasets. For datasets with a temporal response rather than a single scalar per word, the ridge encoder fits a lag-matrix TRF: one multi-output ridge regression predicts every time lag jointly from the same predictors, rather than one regression per lag. Broderick's EEG is downsampled to 32 Hz and epoched from -0.2 to 1.0 s relative to word onset (38 lags); Podcast's ECoG uses the same 32 Hz rate over a wider -1.0 to 1.0 s window. Fedorenko2016 has no temporal axis: its intracranial recordings are already reduced to a single scalar response per (word, electrode) pair.

Podcast's per-region mixed-effects model. For the surprisal track, Podcast is fit with a linear mixed-effects model on the response averaged over the electrodes of each language region (Appendix G), with a random intercept per subject and no electrode term. One model is fit at each lag of the 0–600 ms post-onset window (19 lags), and the score is the mean over lags rather than the value at a peak lag, which avoids selecting the lag on held-out data. The matching reference is computed lag by lag and region by region with the split-half procedure of Appendix A.2 and averaged over lags in the same way; each region's score is normalized by its own reference before the regions are combined (weighted by held-out observations). Held-out partitions are formed from contiguous 30-word chunks of the transcript.

fMRI encoding pipelines. For Le Petit Prince and Natural Stories fMRI, word-level predictors are convolved with a canonical double-gamma HRF (Glover, 1999), implemented via `nilearn`, before ridge or mixed-effects fitting; both use $\mathrm{TR} = 2$ s and discard the first and last volumes of each run for signal equilibration. Le Petit Prince additionally restricts voxels to the top 25% by split-half reliability before averaging into ten bilateral parcels of the Fedorenko language-network atlas, and its ridge analysis is computed on the group-average BOLD signal rather than per subject, following Bonnasse-Gahot & Pallier (2024). Natural Stories fMRI aggregates BOLD into 26 functional regions of interest (6 left-hemisphere language-network parcels and 20 bilateral multiple-demand

parcels) following Shain et al. (2022). Tuckute's sentence-level design uses only the 1,000-sentence naturalistic condition; the adversarially optimized conditions designed to drive or suppress the language network are not used, since only 2–3 of the 10 participants saw each of those sentences. Pereira's surprisal analysis is restricted to the left anterior and posterior temporal language ROIs (Appendix G).

Linear mixed-effects models. All LME models are fit in R via `lme4`. Held-out partitions are formed from contiguous chunks of the data (words, sentences, or short time windows depending on the dataset), shuffled with a fixed seed before assignment to train and held-out splits, so that both partitions sample the full range of each recording or corpus. If a model with the full random-effects structure of Appendix G fails to converge, we follow the simplification procedure of Barr et al. (2013): the correlation between random slopes is dropped first, and if convergence still fails, the slope itself is dropped. The full and nuisance-only models are always compared on the same test observations (Equation 1).

Group analysis. Equation 2 is a second-stage analysis of the final-layer prediction scores, fit with `lme4` and `lmerTest` (restricted maximum likelihood, Satterthwaite degrees of freedom, `bobyqa` optimizer). Model counts in the Pareto plots and regressions differ because the latter also require perplexity, which was not computed for the three OLMo-2 models. Appendix E gives the robustness analyses.

E Robustness of the Group Associations

Table 10 reports the robustness refits that address dependence between models, the flagged likelihood references and the influence of single datasets, and Figure 5 summarizes them. Three further refits of the preferred specification change no conclusion: winsorizing scores at the 5th and 95th percentiles within unit, excluding the six GPT-BERT models, and adding $\log_{10}$ depth as a covariate (all estimates in the released coefficient file). The first column (S0) reproduces a specification with only a dataset-unit random intercept, which treats the 1,822 behavioral and 1,257 neural rows as otherwise independent. For the behavioral constraint contrasts, its standard errors are about four times smaller than those of the preferred specification, which adds random effects for model lineage, model and corpus and unit-specific slopes for size and perplexity: rows from the same model or the same lineage are far from independent. For neural contrasts the two specifications give more similar standard errors, because there the residual variation between units dominates. Several associations that are significant under S0 are not under the preferred specification, including the behavioral memory contrast and the combined contrasts in the embedding track.

The lineage bootstrap resamples lineages with replacement within each constraint group and refits S0, so its uncertainty reflects the variation between independent architectures and training pipelines rather than between rows. Because the constrained groups contain only five or six lineages, its intervals are approximate. Several of the preferred and joint fits are singular, with a lineage or model variance estimated at zero once the other term absorbs it; this does not affect the fixed-effect estimates.

Three patterns are robust to all checks: the behavioral data contrast and the behavioral memory-by-data interaction in the embedding track, and the neural memory contrast in the embedding track. The negative size and perplexity coefficients for behavioral surprisal are also robust. In contrast, the behavioral data contrast in the surprisal track depends on the inclusion of UCL self-paced reading, and the neural surprisal constraint contrasts vanish when the four units with flagged likelihood references are excluded. Figure 6 shows the corresponding per-unit estimates.

F Design-Based Comparisons: Details

Figure 7 shows the scores of the six memory-limited architectures of Bu et al. (2026) by training corpus, and Figure 8 the memory contrasts by mechanism summarized in Table 3. Per-architecture fits show the same picture as the pooled within-architecture model. For behavioral embeddings, no architecture predicts reliably better when trained on BabyLM data than on the Pile, and all six predict reliably worse at either 10M or 100M words (by .005 to .032). For neural embeddings the ar-

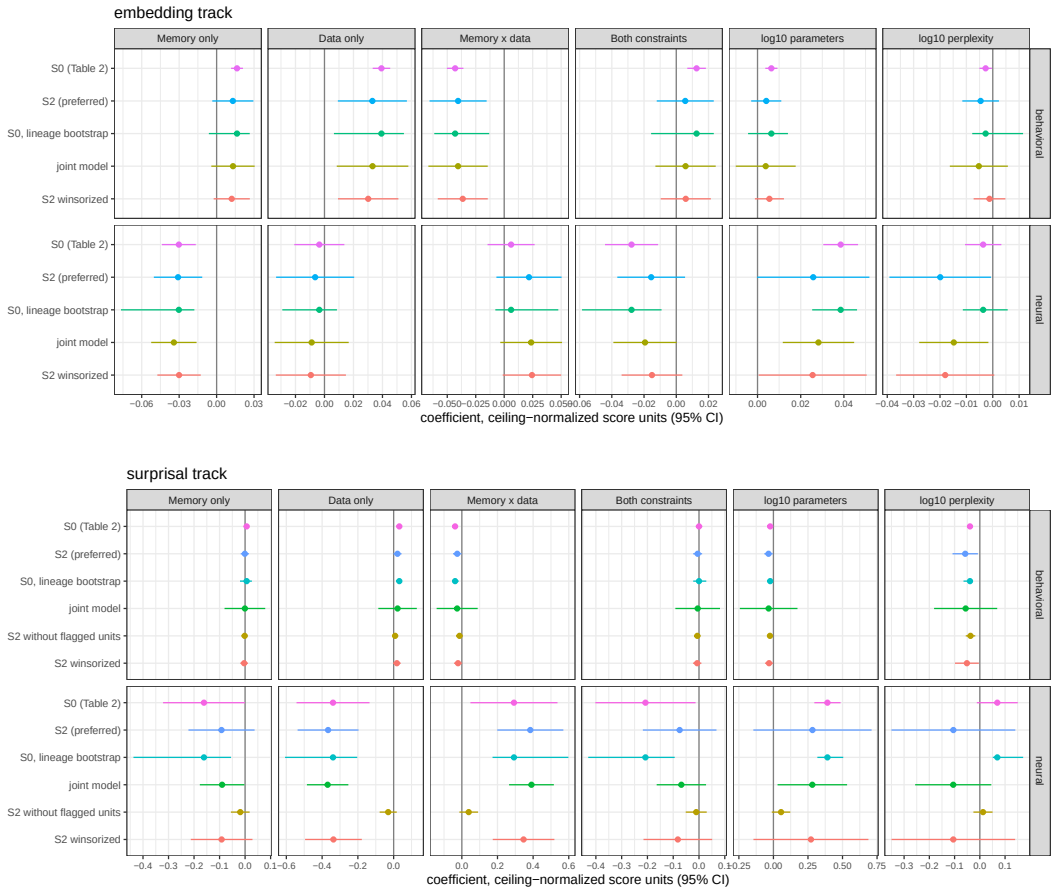


Figure 5: Group associations under the main specifications, embedding track (top) and surprisal track (bottom); points are estimates and bars 95% intervals in ceiling-normalized score units. S0: dataset-unit random intercept only; S2: preferred specification (Table 2); lineage bootstrap: S0 with lineages resampled within constraint groups; joint model: domain-specific effects from the joint behavioral and neural model; the last two rows refit S2 without the units with flagged likelihood references (surprisal only) or with scores winsorized within unit.

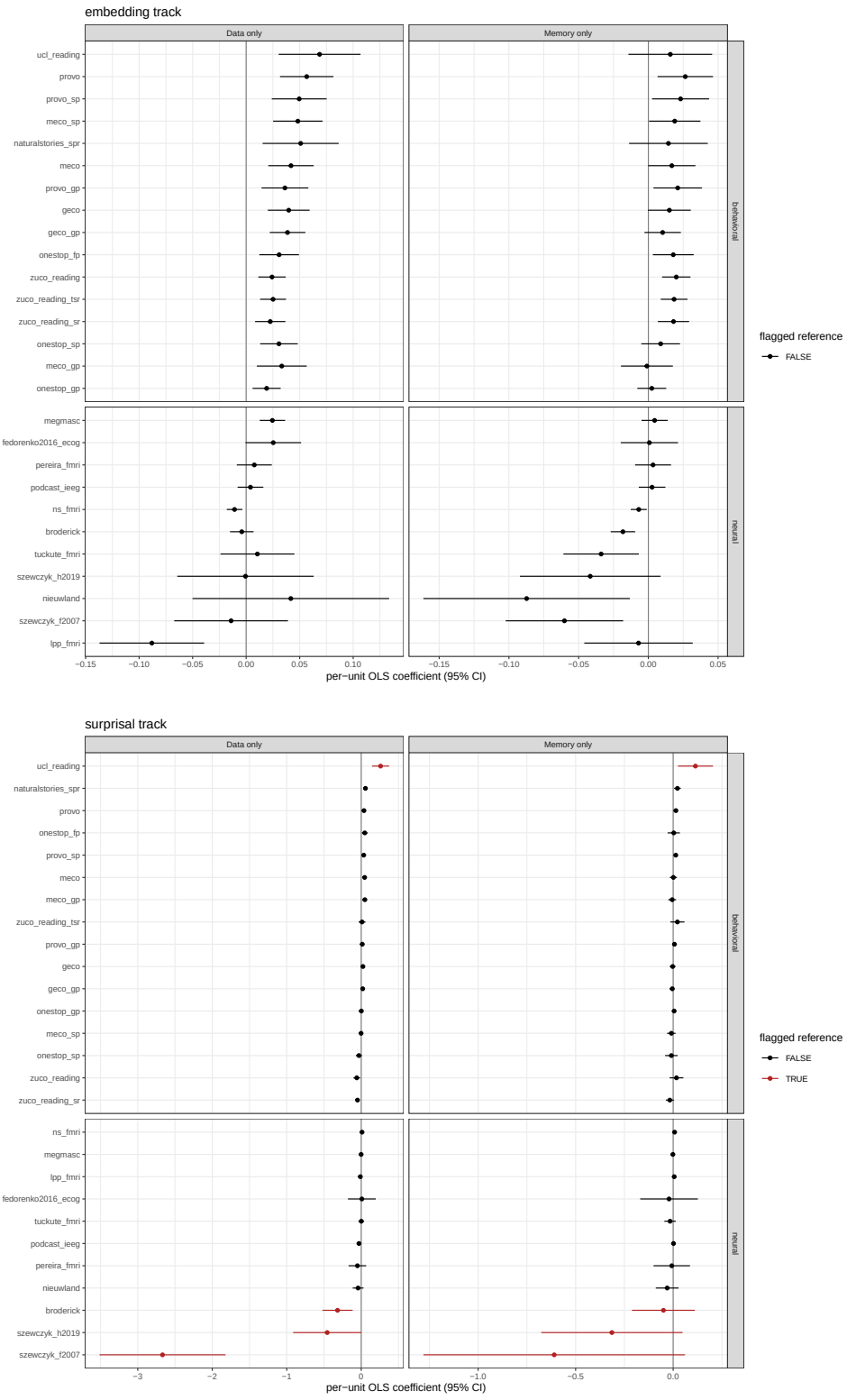


Figure 6: Memory-only and data-only contrasts estimated separately within each dataset unit (ordinary least squares with size and perplexity, 95% intervals), embedding track (top) and surprisal track (bottom). Red: units whose likelihood references are flagged in Table 5. With about 115 models per unit, single-unit estimates are imprecise; the pooled estimates of Table 2 borrow strength across units.

Table 10: Robustness of the constraint contrasts. Cells are coefficients (SE) from separate refits. S0 has a dataset-unit random intercept only and treats rows as otherwise independent; S2 is the preferred specification of Table 2. Lineage bootstrap: S0 refit with model lineages resampled within constraint groups (999 resamples; SE is the bootstrap SD, p two-sided). No † units: S2 without the four units whose likelihood references are flagged in Table 5 (surprisal only). Leave one unit out: range of S2 estimates over refits that each omit one dataset unit, with the number reaching $p < .05$. Refits with winsorized scores, without the GPT-BERT models, or with depth as a covariate change no conclusion (Section E).

Term	S0	S2 (Table 2)	Lineage bootstrap	No † units	Leave one unit out
Embedding, behavioral					
Memory only	$.016\ (.002)^{***}$	$.013\ (.008)$	$.016\ (.008)$	–	.012 to .015 (0/16)
Data only	$.039\ (.003)^{***}$	$.033\ (.012)^{**}$	$.039\ (.013)^{*}$	–	.031 to .035 (16/16)
Memory × data	$-.043\ (.004)^{***}$	$-.040\ (.012)^{**}$	$-.043\ (.013)^{*}$	–	$-.042$ to $-.039$ (16/16)
Both constraints	$.013\ (.003)^{***}$	$.006\ (.009)$	$.013\ (.010)$	–	.004 to .007 (0/16)
Embedding, neural					
Memory only	$-.030\ (.007)^{***}$	$-.031\ (.010)^{**}$	$-.030\ (.015)^{***}$	–	$-.035$ to $-.024$ (11/11)
Data only	$-.004\ (.009)$	$-.006\ (.013)$	$-.004\ (.010)$	–	$-.011$ to $-.001$ (0/11)
Memory × data	$.006\ (.011)$	$.022\ (.014)$	$.006\ (.014)$	–	.013 to .027 (0/11)
Both constraints	$-.028\ (.008)^{***}$	$-.016\ (.011)$	$-.028\ (.012)^{**}$	–	$-.024$ to $-.006$ (1/11)
Surprisal, behavioral					
Memory only	$.006\ (.005)$	$-.001\ (.008)$	$.006\ (.012)$	$-.002\ (.006)$	$-.005$ to .001 (0/16)
Data only	$.032\ (.007)^{***}$	$.022\ (.011)$	$.032\ (.006)^{***}$	$.008\ (.008)$	.008 to .028 (4/16)
Memory × data	$-.038\ (.008)^{***}$	$-.026\ (.012)^{*}$	$-.038\ (.009)^{***}$	$-.013\ (.009)$	$-.032$ to $-.013$ (14/16)
Both constraints	$.000\ (.006)$	$-.005\ (.009)$	$.000\ (.013)$	$-.006\ (.006)$	$-.009$ to $-.002$ (0/16)
Surprisal, neural					
Memory only	$-.161\ (.082)^{*}$	$-.092\ (.065)$	$-.161\ (.096)^{**}$	$-.019\ (.018)$	$-.104$ to $-.052$ (0/11)
Data only	$-.339\ (.104)^{**}$	$-.367\ (.084)^{***}$	$-.339\ (.103)^{***}$	$-.030\ (.024)$	$-.407$ to $-.122$ (11/11)
Memory × data	$.292\ (.125)^{*}$	$.384\ (.092)^{***}$	$.292\ (.109)^{***}$	$.039\ (.026)$	.150 to .425 (11/11)
Both constraints	$-.208\ (.099)^{*}$	$-.075\ (.072)$	$-.208\ (.089)^{***}$	$-.011\ (.020)$	$-.086$ to $-.035$ (0/11)

chitectures disagree: the forgetting architecture improves as data decrease (.035 at 100M and .049 at 10M words), whereas stick-breaking attention is worse at 10M words ($-.059$), so the pooled neural dose-response is not consistent across architectures. These architectures have no softmax-attention counterpart trained on the same three corpora, so the within-architecture comparison isolates the amount of data but cannot test the memory mechanism itself; the Forgetting Transformer pairs provide that test for one mechanism. GPT-BERT models, which combine causal and masked objectives, are evaluated through their causal language-modeling head, so their surprisal is a left-to-right next-token prediction; excluding them does not change the group contrasts (Appendix E).

G Linear Mixed-Effects Formulas

This appendix specifies the linear mixed-effects (LME) models underlying the surprisal track analyses of Section 3.4, in standard `lme4` formula notation. All results reported in the paper use the final layer: for each language model and dataset we fit a *baseline* model containing only nuisance covariates, and a *full* model that additionally includes the model's surprisal, with the two compared via the log-likelihood difference described in Appendix A.2. All continuous predictors are z-scored within dataset; `surprisal_z` denotes the next-word surprisal of the evaluated model (summed over the word's subword tokens), and `prev_surprisal_z` the same quantity for the preceding word (a spillover control).

G.1 Self-paced reading and eye-tracking

Reading times and fixation durations across Natural Stories, UCL Reading, GECO, MECO, Provo, OneStop, and ZuCo Reading are modeled with a common specification, with word length, frequency, and sentence position as nuisance covariates and by-subject random slopes:

```
baseline: fdur ~ sentpos_z + wlen_z + freq_z +
                 (1 + sentpos_z + wlen_z | subject)
full: fdur ~ sentpos_z + wlen_z + freq_z + surprisal_z
```

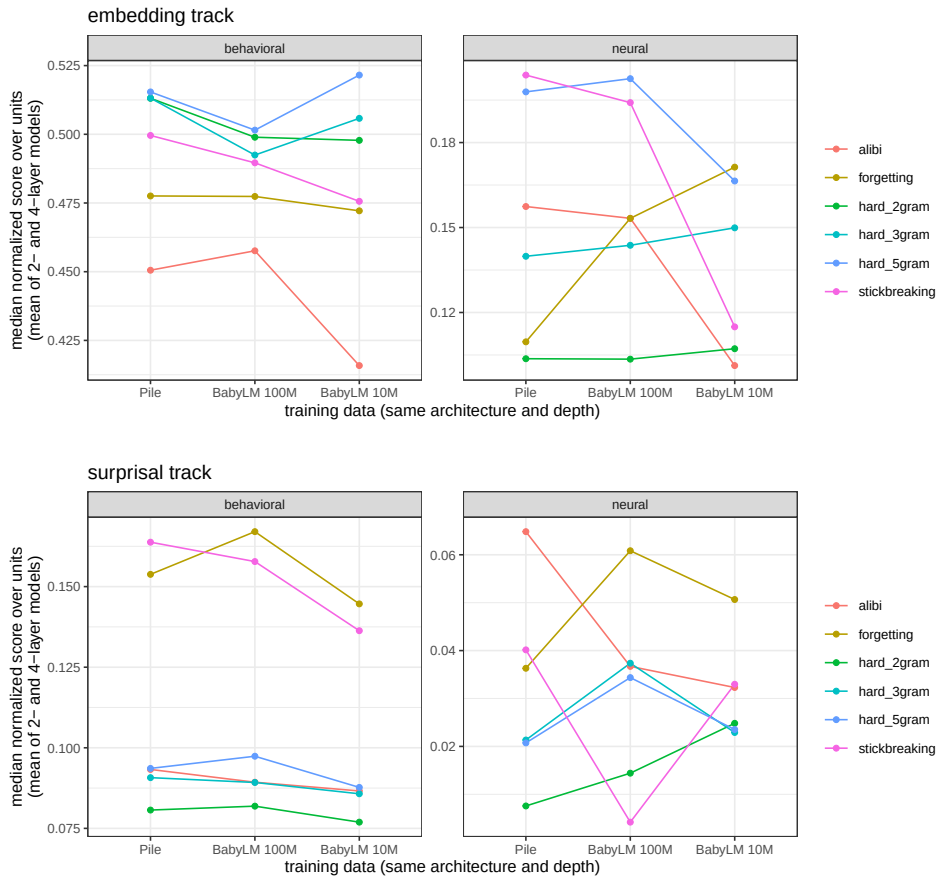


Figure 7: Within-architecture data comparison, embedding track (top) and surprisal track (bottom). Each line is one memory-limited architecture; points are the median normalized score over dataset units, averaged over the 2- and 4-layer models trained on each corpus.

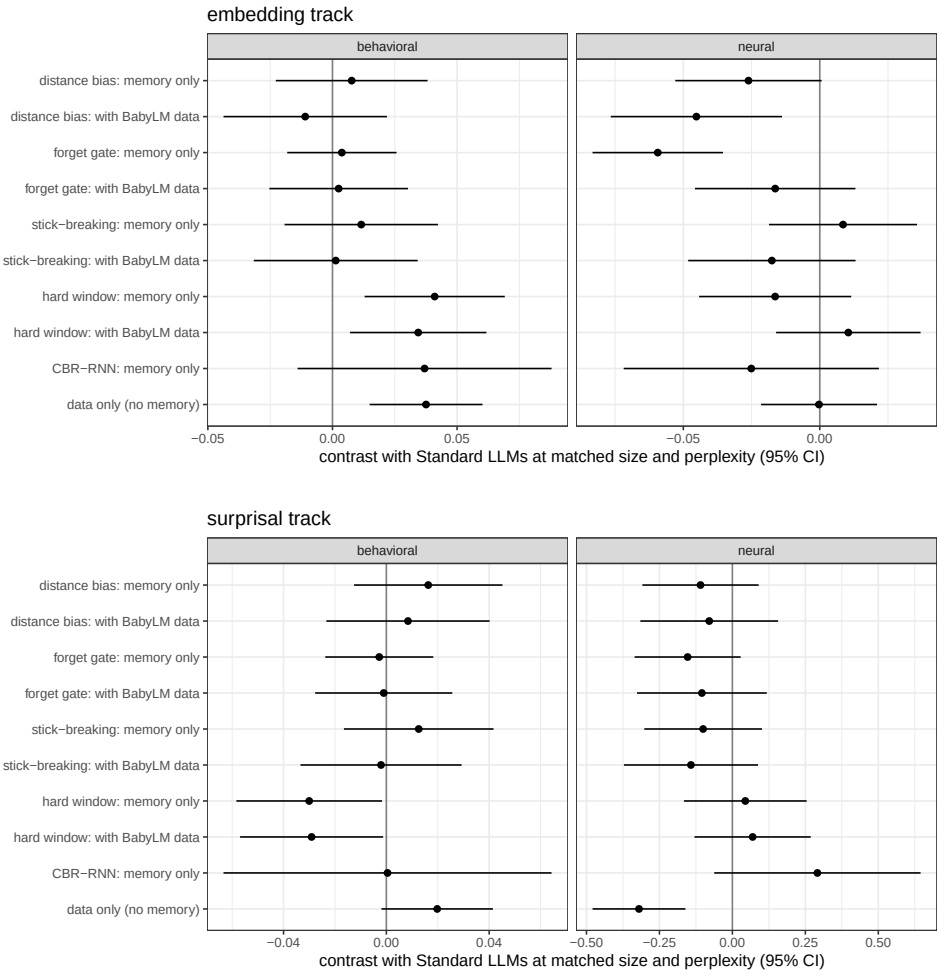


Figure 8: Memory contrasts by mechanism (preferred specification), embedding track (top) and surprisal track (bottom): memory-only contrasts and, for mechanisms also trained on BabyLM data, the combined contrast with Standard LLMs; the last row is the data-only contrast. 95% intervals.

```
+ prev_surprisal_z +
              (1 + sentpos_z + wlen_z + surprisal_z +
prev_surprisal_z | subject)
```

G.2 N400 EVENT-RELATED POTENTIALS

Nieuwland is modeled following the specification of its originating study rather than our shared self-paced-reading/eye-tracking structure, since its multi-site design calls for a different random-effects treatment. Following Nieuwland et al. (2018), laboratory is entered as a fixed rather than random effect, since a random-effect variance is not reliably estimable from only nine sites; there is no electrode term, as N400 amplitude is already averaged across centro-parietal channels in this dataset, nor a frequency covariate, matching the original analysis:

```
baseline: n400 ~ lab + (1 | subject) + (1 | item)
full: n400 ~ lab + surprisal_z + (1 | subject) + (1 |
item)
```

Federmeier et al. (2007) and Hubbard et al. (2019) are fit separately, one model per study, with the specification of the analysis scripts of the Szewczyk and Federmeier collection (Szewczyk & Federmeier, 2022). N400 amplitude is pre-averaged over four centro-parietal electrode clusters into one value per trial, so there is no electrode term. The nuisance covariates are the pre-stimulus baseline amplitude (`bline_z`), log frequency, word position in the sentence, orthographic neighbourhood (`old20_z`) and concreteness, and random slopes are uncorrelated (`||`); `item` is the sentence frame:

```
baseline: n400 ~ bline_z + freq_z + pos_start_z +
old20_z + concr_z +
                 (1 + bline_z + pos_start_z || subject)
+ (1 + bline_z || item)
full: n400 ~ bline_z + freq_z + pos_start_z + old20_z +
concr_z + surprisal_z +
              (1 + bline_z + pos_start_z || subject) +
(1 + bline_z || item)
```

G.3 CONTINUOUS EEG AND MEG

For Broderick, one model is fit on the data of all subjects pooled, with a random intercept for subject. `region` denotes four anatomical scalp quadrants (left, right, posterior, occipital), the same for every subject.

```
baseline: eeg ~ wlen_z + freq_z + (1 | subject) + (1 |
region)
full: eeg ~ wlen_z + freq_z + surprisal_z +
prev_surprisal_z + (1 | subject) + (1 + surprisal_z
| region)
```

For MEG-MASC, the response is the mean amplitude in a 250–450 ms post-onset window, averaged within six sensor groups (`region`, the same grouping for every subject); one pooled model is fit over all subjects, with a random intercept for subject. The nuisance covariates are word duration, word length, frequency and the duration of the previous word:

```
baseline: y ~ worddur_z + wlen_z + freq_z +
prev_worddur_z + (1 | subject) + (1 | region)
full: y ~ worddur_z + wlen_z + freq_z + prev_worddur_z +
surprisal_z + prev_surprisal_z +
           (1 | subject) + (1 + surprisal_z | region)
```

G.4 FMRI

For Natural Stories fMRI, `story` is a crossed random intercept (there are only eight stories) and the single random slope is on `subject`; no spillover term is included, since HRF convolution already

blends the contributions of adjacent words into a single BOLD sample. Dropping the frequency covariate from both models gives the variant without frequency.

```
baseline: bold ~ wlen_z + freq_z + sentpos_z + (1 |
subject) + (1 | story)
full: bold ~ wlen_z + freq_z + sentpos_z + surprisal_z
+
             (1 + surprisal_z | subject) + (1 | story)
```

For Le Petit Prince, one model is fit on the BOLD signal of all subjects; `region` denotes one of ten bilateral language-network parcels from the Fedorenko functional atlas, `section` one of the nine narrative chapters, and `subject` enters as a random intercept.

```
baseline: bold ~ wlen_z + freq_z + sentpos_z + (1 |
region) + (1 | section) + (1 | subject)
full: bold ~ wlen_z + freq_z + sentpos_z + surprisal_z
+
             (1 + surprisal_z | region) + (1 |
section) + (1 | subject)
```

For Tuckute, the response is a single BOLD scalar per (subject, ROI, sentence), so word length and frequency are averaged over the sentence (`mean_wlen_z`, `mean_freq_z`) and surprisal is computed over the whole sentence rather than word-by-word. The model is fit on the naturalistic sentence condition only.

```
baseline: y ~ mean_wlen_z + mean_freq_z +
              (1 + mean_wlen_z + mean_freq_z | subject)
+ (1 | roi) + (1 | sentence_id)
full: y ~ mean_wlen_z + mean_freq_z + surprisal_z +
          (1 + mean_wlen_z + mean_freq_z + surprisal_z |
subject) + (1 | roi) +
          (1 + surprisal_z | sentence_id)
```

For Pereira, which uses the same per-sentence design as Tuckute but with sentences of variable length, we follow the specification of Lin & Schuler (2026): sentence length (`sentlen_z`) is retained as an additional nuisance covariate alongside mean word length and frequency, and surprisal is again computed over the whole sentence rather than word-by-word. Only two regions of the language-atlas parcellation, the left anterior and left posterior temporal ROIs, carry a reliable sentence-level signal in the split-half ΔLL ceiling (Appendix A.2); all other ROIs are at noise level. We therefore fit the model separately within each of these two ROIs, so the `roi` random effect is dropped, normalize each ROI's held-out ΔLL by its own ceiling, and average the two normalized scores.

```
baseline: y ~ sentlen_z + mean_wlen_z + mean_freq_z +
              (1 + sentlen_z + mean_wlen_z + mean_freq_z
| subject) + (1 | sentence_id)
full: y ~ sentlen_z + mean_wlen_z + mean_freq_z +
surprisal_z +
          (1 + sentlen_z + mean_wlen_z + mean_freq_z +
surprisal_z | subject) +
          (1 + surprisal_z | sentence_id)
```

G.5 Intracranial recordings

For Fedorenko2016, electrodes are patient-specific, so `electrode` is nested within `subject` rather than treated as a crossed effect. Podcast is instead fit per region on region-averaged electrodes, so no electrode term is needed: for each language region (anterior temporal, inferior frontal, posterior temporal; left-hemisphere electrodes assigned to the Fedorenko language parcels, with the orbital part of the inferior frontal gyrus merged into the inferior frontal region and the middle frontal gyrus dropped because it has fewer than four subjects), the high-gamma response is averaged over

the region's electrodes within each subject and word, and the model is fit separately at each lag of the 0–600 ms post-onset window (19 lags on the 31.25 ms grid), at the final layer; no spillover term is included. Responses are standardized with statistics computed on the training chunks only (per electrode, then per subject and region after averaging) and applied unchanged to the held-out chunks, so no information from the held-out data enters the scaling. The headline score is the mean over lags of the held-out ΔLL per observation of each region, divided by that region's own noise ceiling and then averaged across regions weighted by their number of held-out observations.

```
baseline: y ~ wlen_z + freq_z + (1 | subject)
full: y ~ wlen_z + freq_z + surprisal_z + (1 | subject)
```

For Fedorenko2016, spillover is included, along with a richer by-subject random-slope structure:

```
baseline: y ~ wlen_z + freq_z + (1 + wlen_z + freq_z |
subject) +
              (1 | subject:electrode)
full: y ~ wlen_z + freq_z + surprisal_z +
prev_surprisal_z +
          (1 + wlen_z + freq_z + surprisal_z +
prev_surprisal_z | subject) +
          (1 + surprisal_z + prev_surprisal_z |
subject:electrode)
```

H Representational Evaluation

The ridge encoders of the embedding track have a number of free parameters that grows with the hidden size d of the model, so a score that increases with model size may reflect the capacity of the regression rather than a better representation. Representational similarity analysis (RSA; Kriegeskorte et al., 2008) removes this dependence: whatever d is, a model yields one stimulus-by-stimulus dissimilarity matrix (RDM), and the downstream score involves at most a single model coefficient, with no per-feature weights and no tuned hyperparameter. We run it for both tracks, on the same models and stimuli as the main benchmark and with the final layer of each model. The subsections below describe the dissimilarity matrices, the unfitted RSA score, the confound-controlled RDM regression with its cross-validation, the noise ceilings, and the aggregation.

H.1 Dissimilarity matrices

For each stimulus i, let h_i denote its final-layer embedding and s_i its surprisal. Embedding features are standardized across the scored stimuli, $\widetilde{h}_{ik} = (h_{ik} - \mu_k)/\sigma_k$, with constant features set to zero. This preprocessing addresses unequal feature scales (Timkey & van Schijndel, 2021); it is distinct from standardizing outcome scores across models. Because the statistics include all scored stimuli, it is a transductive transformation, although it uses no response values. Raw last-layer states of Transformers contain a handful of high-magnitude dimensions that dominate cosine and correlation distances (Timkey & van Schijndel, 2021), which is why every feature is standardized before distances are computed. Surprisal s_i is the surprisal of the (critical) word, or the summed surprisal of the sentence for sentence-level datasets, exactly as used by the mixed-effects analyses. Model distances are

$$D^E_{ij} = 1 - \mathrm{corr}(\widetilde{h}_i, \widetilde{h}_j), \qquad D^S_{ij} = |s_i - s_j|.$$

Behavioral distances are absolute differences in item-mean reading responses. Scalar N400 responses are treated analogously; multivariate neural response patterns use correlation distance. Stimulus alignment and response aggregation are dataset-specific. The current RSA Pareto analysis uses language-region units for Pereira and Tuckute, language parcels for Podcast, regional responses for Broderick and MEG-MASC, and the three N400 datasets.

All stimulus pairs are used where feasible; GECO and OneStop use a fixed random sample of 30 million pairs, the same for every model. Model, response, nuisance, and ceiling quantities use corresponding pairs within each analysis. The per-dataset correlations of Table 13 use each dataset's full response RDM rather than the regional units of the Pareto composites, so their values are not directly comparable with those composites.

H.2 Unfitted Pearson RSA

Let $v(D)$ denote the retained upper-triangle entries. Unfitted RSA is

$$R = \mathrm{corr}\big(v(D^{\mathrm{model}}), v(D^{\mathrm{response}})\big).$$

It learns no weights mapping model features to responses and does not remove lexical confounds. The neural RSA Pareto score divides each response unit's RSA by its own positive RSA ceiling, summarizes these ratios by the median within a dataset, and takes the median across the eight neural datasets. Behavioral composites take the median across the 16 measure-specific ratios. All 118 plotted models have complete coverage at these dataset/measure levels. The RSA ceiling uses the square root of Spearman–Brown-corrected split-half RDM reliability.

H.3 Confound-controlled RDM regression

For each retained pair, the nuisance model predicts response distance from an intercept and absolute differences in word length and frequency. The full model adds one model-distance predictor:

$$D_{ij}^{\mathrm{response}} = a + b_L D_{ij}^{\mathrm{length}} + b_F D_{ij}^{\mathrm{frequency}} + e_{ij},$$
$$D_{ij}^{\mathrm{response}} = a + b_L D_{ij}^{\mathrm{length}} + b_F D_{ij}^{\mathrm{frequency}} + b_M D_{ij}^{\mathrm{model}} + e_{ij}.$$

For Federmeier et al. (2007) and Hubbard et al. (2019), the nuisance model additionally includes absolute differences in word position, orthographic neighborhood (OLD20) and concreteness, to match the covariates of their published mixed-effects formulas; for sentence datasets, the lexical distances are computed from sentence length, mean word length and mean word frequency. Random-effect structures have no counterpart on an item-averaged target and are not reproduced. All distance entries are z-scored before fitting. The models are fit by ordinary least squares, and Gaussian test log likelihood uses the variance estimated from training residuals. The score is the full-minus-baseline test log likelihood per pair. There is one model-distance coefficient regardless of embedding width; no feature reweighting is fitted.

Cross-validation partitions stimuli. For a held-out fold, training pairs have both stimuli outside the fold and test pairs have both inside it; pairs crossing the boundary are omitted. This prevents training and test pairs from sharing stimuli in a split (Schütt et al., 2023). Following Schütt et al. (2023), the number of folds depends on the number of stimuli n: $K = 2$ for $n < 12$, 3 for $12 \le n < 24$, 4 for $24 \le n < 40$, and 5 for $n \ge 40$. Test log likelihoods are summed over folds, and the procedure is repeated with new random partitions (10 by default), the reported score being the mean over repeats. Partitions are fixed per dataset and unit, so all models and both tracks are scored on identical train and test pairs, and base and full models are always compared on the same test pairs. Pairs sharing a stimulus remain correlated within the training and test sets, which violates the independence assumption of the Gaussian likelihood and inflates absolute values by roughly the same amount for all models. The pair likelihood is therefore used to rank models, not as a source of independent-pair significance tests, and its absolute scale is not interpreted. Negative scores indicate worse held-out fit than the nuisance baseline.

Figure 4 uses dataset- and measure-level RDM-regression scores, divided by the matching split-half likelihood references and then aggregated by medians. This differs from the unit-wise Pearson RSA normalization described above. In both cases, only usable positive references enter the composite; negative model scores and ratios above one are retained. Normalization does not make the direct-prediction, Pearson RSA, and RDM-likelihood scales interchangeable.

H.4 Noise ceilings for representational scores

Both representational scores are expressed relative to the reliability of the human data, computed in the same RDM space, with the same item folds, nuisance RDMs and pair likelihood as the models. Participants are randomly split into two equal halves A and B (15 splits); items must be observed in both halves. For neural datasets the split is over subjects and is carried out separately for each response unit (ROI, channel, region, or region $\times$ lag), so each unit has its own ceiling.

RSA-correlation ceiling. The split-half reliability r_{half} is the Pearson correlation between the entries of the two half-sample response RDMs, averaged over splits and extrapolated by the Spearman–

Brown formula; the ceiling for a correlation is $\sqrt{r_{\mathrm{SB}}}$, as in Appendix A.1. Units whose r_{half} is not positive have no defined ceiling and are excluded.

Split-half $\Delta\ell$ reference. Half A's item-mean RDM is used as the predictor of half B's item-mean RDM in the full model above, in place of a model RDM, and compared with the same nuisance-only baseline on the same test pairs; the roles of A and B are then swapped, the two totals are summed and divided by the total number of test pairs, and the reference is the mean over splits. The predictor is built from only half of the participants, so the reference is conservative and no Spearman–Brown correction is available on the likelihood scale; ratios above 1 can occur.

Table 11 reports the RSA-correlation references with the number of usable units, and Table 12 the RDM-regression references used in Figure 4.

Table 11: RSA-correlation references and maximum final-layer normalized scores. For neural datasets, Reference is the median of the unit-level ceilings, reported for context; normalization is performed within units before taking the median and is not division by this displayed summary. Units counts usable positive-ceiling response units. For behavioral measures there is one ceiling per measure. Maximum columns are maxima across models of the dataset/measure score.

Dataset / measure	Reference	Units	Max. emb.	Max. sur.
Pereira fMRI	0.260	20	0.152	0.058
Tuckute fMRI	0.288	13	0.072	0.176
Podcast ECoG	0.224	122	0.050	0.059
Broderick EEG	0.105	4	0.020	0.111
MEG-MASC	0.154	6	0.018	0.047
Nieuwland N400	0.787	1	0.118	0.185
Federmeier 2007 N400	0.363	1	0.171	0.649
Hubbard 2019 N400	0.517	1	0.204	0.554
Provo / FP	0.863	1	0.130	0.458
Provo / GP	0.735	1	0.049	0.252
Provo / SP	0.884	1	0.129	0.460
MECO / FP	0.842	1	0.139	0.425
MECO / GP	0.487	1	0.043	0.125
MECO / SP	0.899	1	0.147	0.456
UCL / SPR	0.505	1	0.017	0.177
ZuCo / NR	0.766	1	0.125	0.555
ZuCo / SR	0.799	1	0.104	0.501
ZuCo / TSR	0.555	1	0.099	0.361
Natural Stories / SPR	0.873	1	0.033	0.254
GECO / GP	0.296	1	0.031	0.117
GECO / FP	0.420	1	0.067	0.272
OneStop / GP	0.526	1	0.022	0.136
OneStop / SP	0.666	1	0.073	0.415
OneStop / FP	0.566	1	0.054	0.419

Table 12: Dataset/measure-level split-half likelihood references for the RDM-regression fronts in Figure 4, in $\Delta\ell$ per pair. They differ from the unit-wise normalization of the RSA correlations. All listed references are positive.

Dataset / measure	Reference	Included
Pereira fMRI	2.98×10^{-4}	Yes
Tuckute fMRI	3.53×10^{-4}	Yes
Broderick EEG	3.28×10^{-5}	Yes
MEG-MASC	8.08×10^{-5}	Yes
Podcast ECoG	3.46×10^{-4}	Yes
Nieuwland N400	1.11×10^{-1}	Yes
Federmeier 2007 N400	1.95×10^{-3}	Yes
Hubbard 2019 N400	9.01×10^{-3}	Yes

Continued on next page

Table 12 – continued

Dataset / measure	Reference	Included
Provo / FP	1.69×10^{-1}	Yes
Provo / GP	7.36×10^{-2}	Yes
Provo / SP	2.13×10^{-1}	Yes
MECO / FP	1.42×10^{-1}	Yes
MECO / GP	7.13×10^{-3}	Yes
MECO / SP	2.67×10^{-1}	Yes
GECO / FP	3.64×10^{-3}	Yes
GECO / GP	8.60×10^{-4}	Yes
OneStop / FP	1.25×10^{-2}	Yes
OneStop / GP	1.22×10^{-2}	Yes
OneStop / SP	3.09×10^{-2}	Yes
Natural Stories / SPR	2.33×10^{-1}	Yes
UCL / SPR	9.74×10^{-3}	Yes
ZuCo / NR	7.40×10^{-2}	Yes
ZuCo / SR	9.17×10^{-2}	Yes
ZuCo / TSR	1.34×10^{-2}	Yes

H.5 Scaling and agreement with direct prediction

Table 13 reports descriptive correlations across the 118 models, computed from the per-model RSA and direct-prediction scores with the current group assignments. The size columns correlate RSA with $\log_{10}$ parameter count, pooled over all models and within constraint groups (the partial correlation controlling for group membership). The agreement columns correlate embedding RSA with direct embedding Pearson scores, and surprisal RSA with direct surprisal $\Delta\ell$ scores. These correlations are not adjusted for model family and do not test a difference between behavioral and neural correlations.

Table 13: Descriptive Pearson correlations across 118 models: RSA with $\log_{10}$ parameter count (size), the same correlation within constraint groups (partial correlation controlling for group membership, so that it does not merely reflect that the smallest models are the constrained ones), and RSA with the corresponding direct-prediction score (agreement: embedding RSA with encoding r, surprisal RSA with surprisal $\Delta\ell$). Emb. denotes embeddings; Sur. denotes surprisal. No model-family adjustment or significance test is applied.

	Size		Size within group		Agreement	
Dataset / measure	Emb.	Sur.	Emb.	Sur.	Emb.	Sur.
Pereira / fMRI	0.698	0.177	0.460	0.056	0.420	0.019
Tuckute B / fMRI	0.437	0.603	0.165	0.302	0.820	0.564
Broderick / EEG	0.606	-0.070	0.371	0.267	0.733	-0.119
MEG-MASC / MEG	-0.558	0.550	-0.444	0.305	0.275	0.138
Nieuwland / N400	0.717	0.729	0.526	0.629	0.532	0.749
Federmeier et al. (2007) / N400	0.884	0.795	0.762	0.558	0.929	0.751
Hubbard et al. (2019) / N400	0.729	0.739	0.545	0.569	0.823	0.517
Podcast / ECoG	0.475	0.131	0.365	-0.127	0.769	0.442
Provo / FP	-0.494	-0.397	-0.183	-0.311	0.678	0.354
Provo / GP	-0.259	-0.527	-0.020	-0.294	0.663	0.017
Provo / SP	-0.498	-0.037	-0.192	-0.113	0.649	0.243
MECO English / FP	-0.597	-0.339	-0.256	-0.214	0.154	0.584
MECO English / GP	-0.394	-0.381	-0.044	-0.266	0.087	-0.194
MECO English / SP	-0.582	-0.058	-0.228	-0.067	0.117	0.227
GECO / FP	-0.366	-0.397	-0.082	-0.240	0.023	-0.272
GECO / GP	-0.133	-0.669	0.018	-0.438	-0.359	-0.304
OneStop / FP	-0.519	0.186	-0.234	0.135	-0.342	0.284
OneStop / GP	0.153	-0.813	0.107	-0.654	0.031	-0.206
OneStop / SP	-0.457	0.562	-0.142	0.336	-0.412	0.751
Natural Stories / SPR	0.101	0.192	0.121	0.202	0.388	0.132
UCL Reading / SPR	-0.043	0.507	0.088	0.418	0.577	0.715
ZuCo / NR	-0.623	0.043	-0.337	0.024	-0.325	0.248
ZuCo / SR	-0.245	0.458	-0.003	0.335	-0.219	0.399
ZuCo / TSR	-0.643	0.545	-0.360	0.378	-0.007	0.694